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(54) COOKING APPARATUS

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CPC ............ *F24C 15/2035* (2013.01); *F24C 7/02* (2013.01); *H05B 6/6414* (2013.01)

(57) ABSTRACT

A cooking apparatus including a main body forming a cooking room therein and including a front plate with an opening to the cooking room; a case positioned to one side of the opening and formed to accommodate a circuit board therein; a door opening and closing the opening and covering a front portion of the case, wherein a portion of a rear surface of the door is depressed inward to accommodate the case. The cooking apparatus includes a circulating air outlet formed at a top of the main body, and configured to filter and discharge air received from below the main body; and a suction grill attachable to and detachable from an upper portion of the case and accommodated in one side of the door, wherein a cooling air inlet, which allows cooling air to enter through the suction grill, is positioned in front of the circulating air outlet.

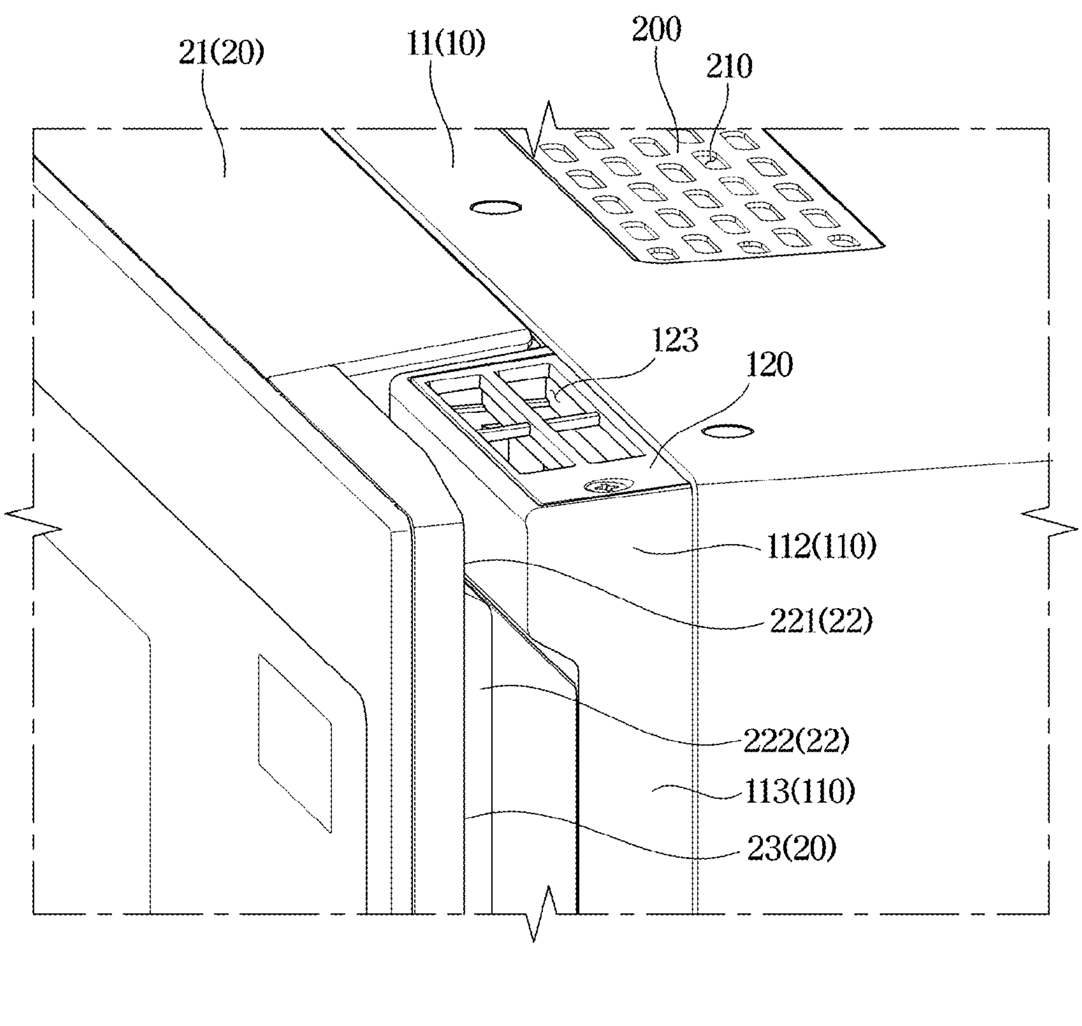


**FIG. 1**

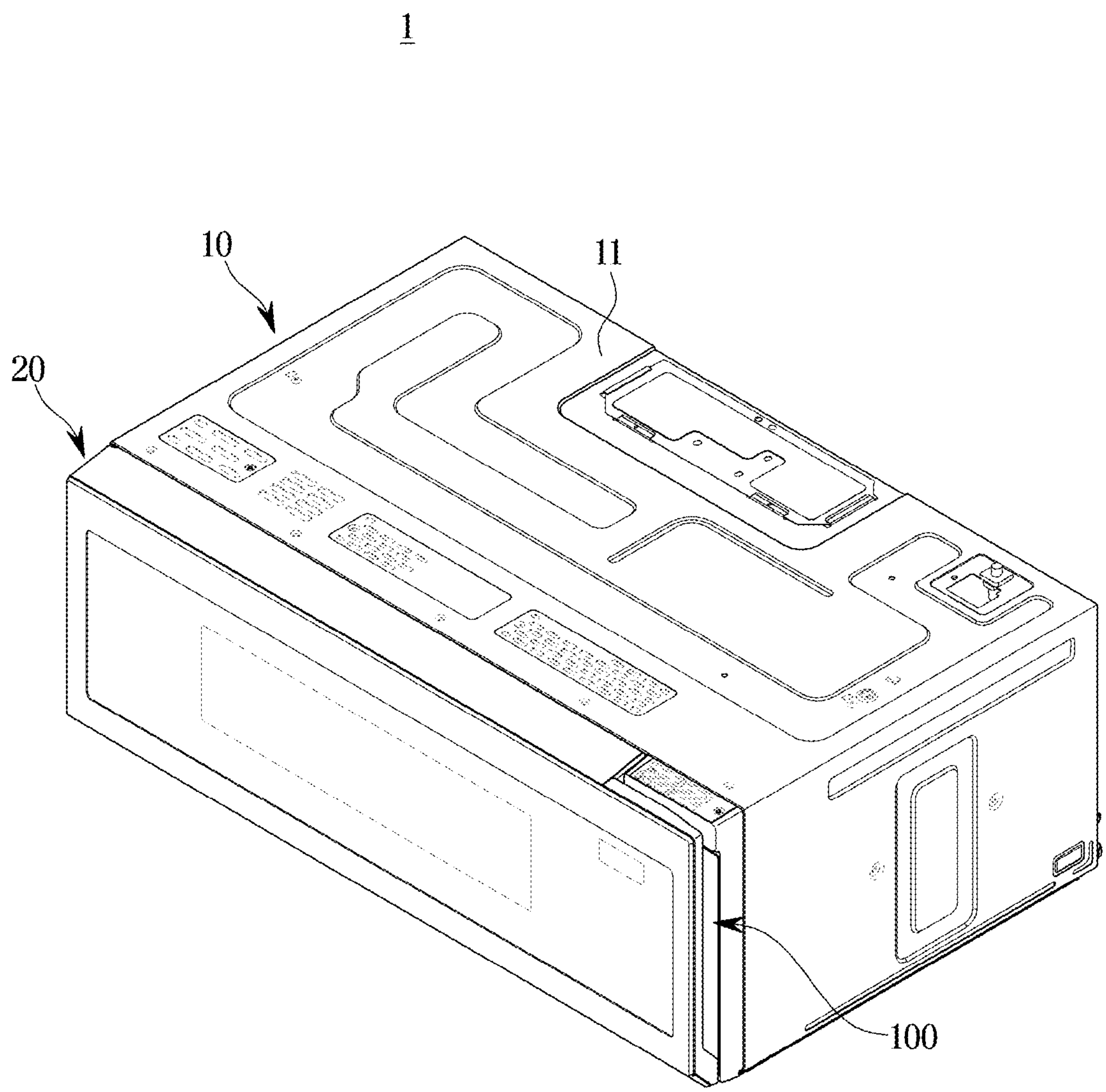


FIG. 2

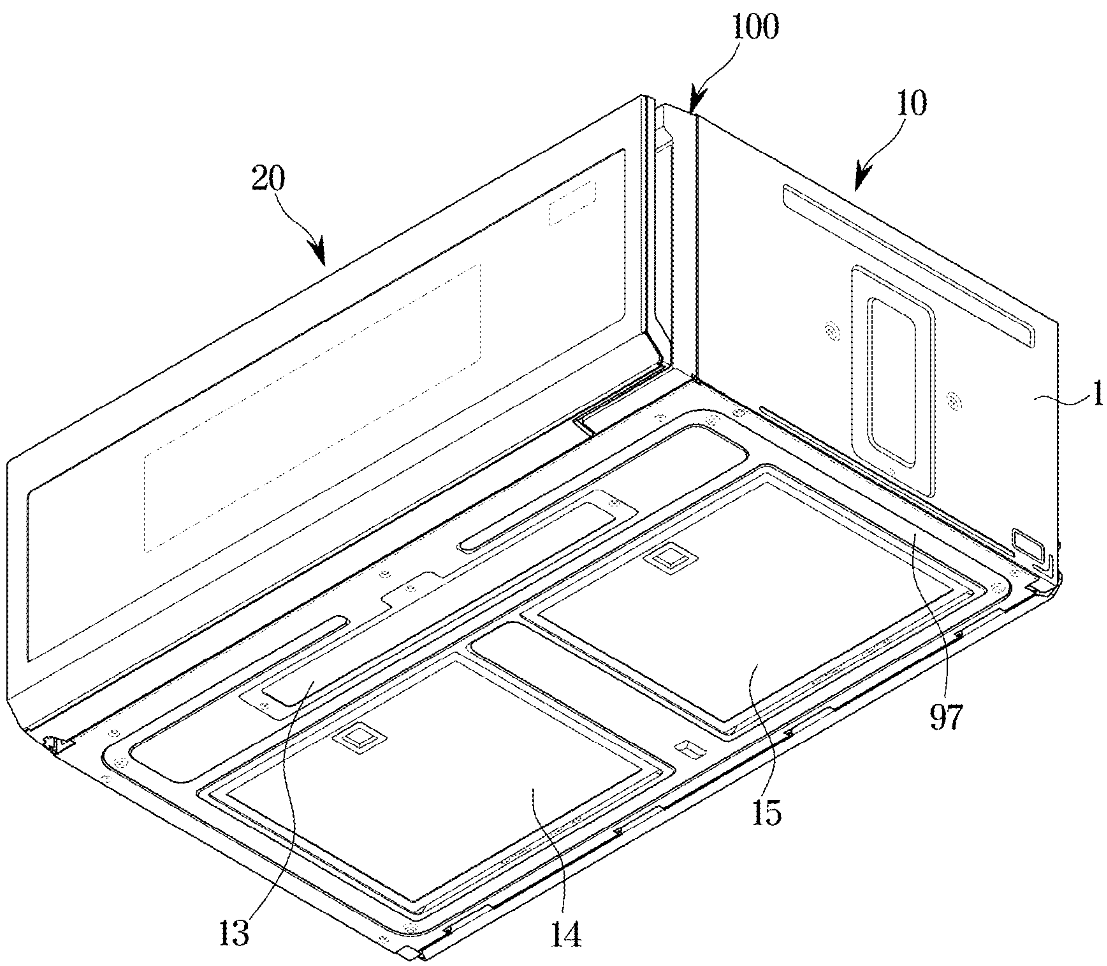


FIG. 3

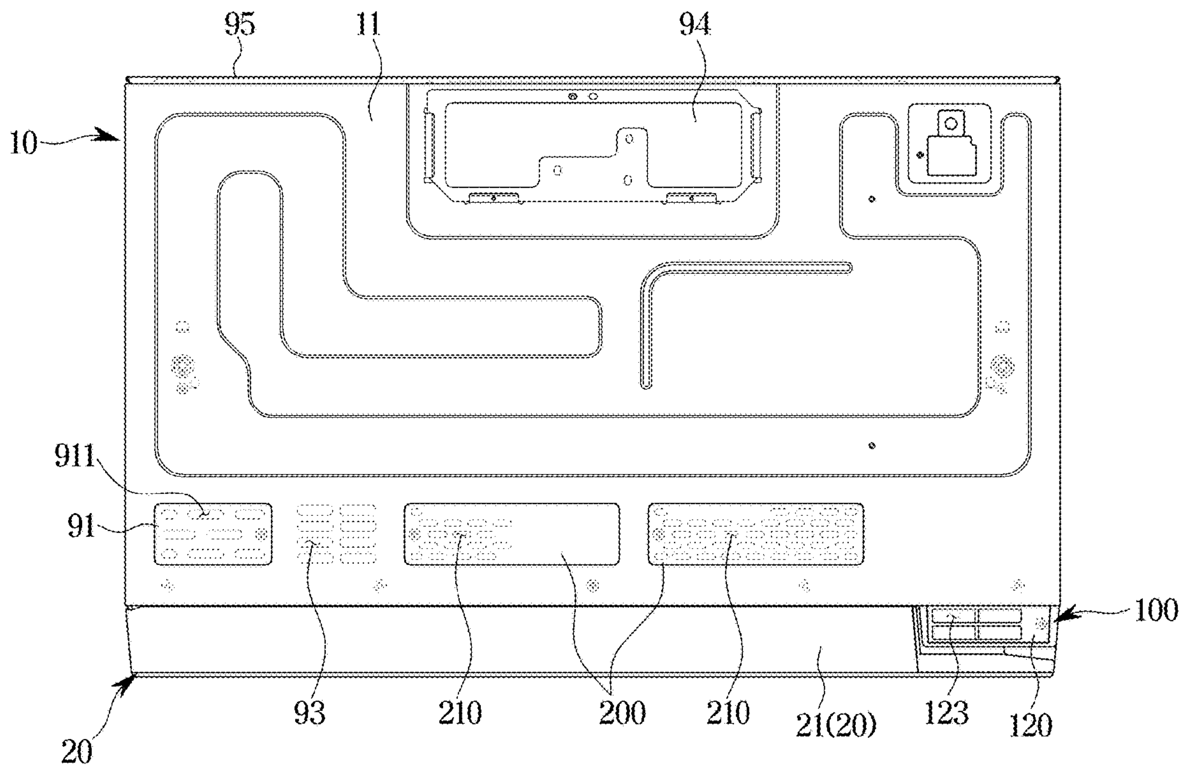


FIG. 4

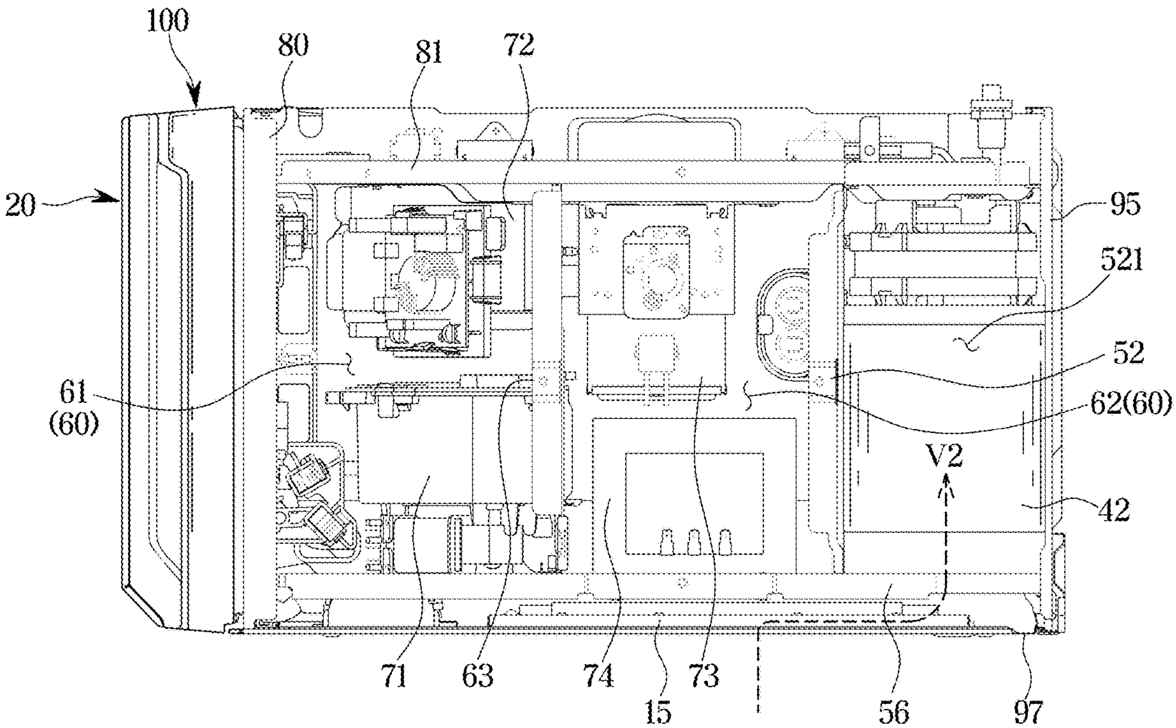


FIG. 5

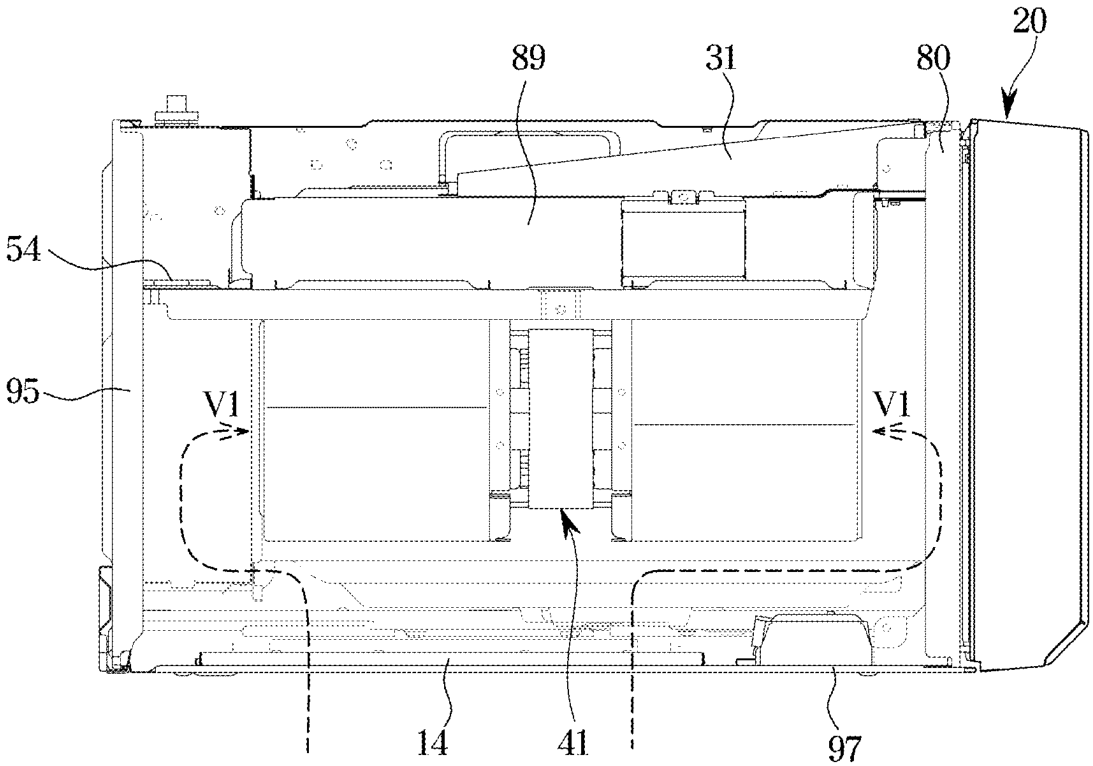


FIG. 6

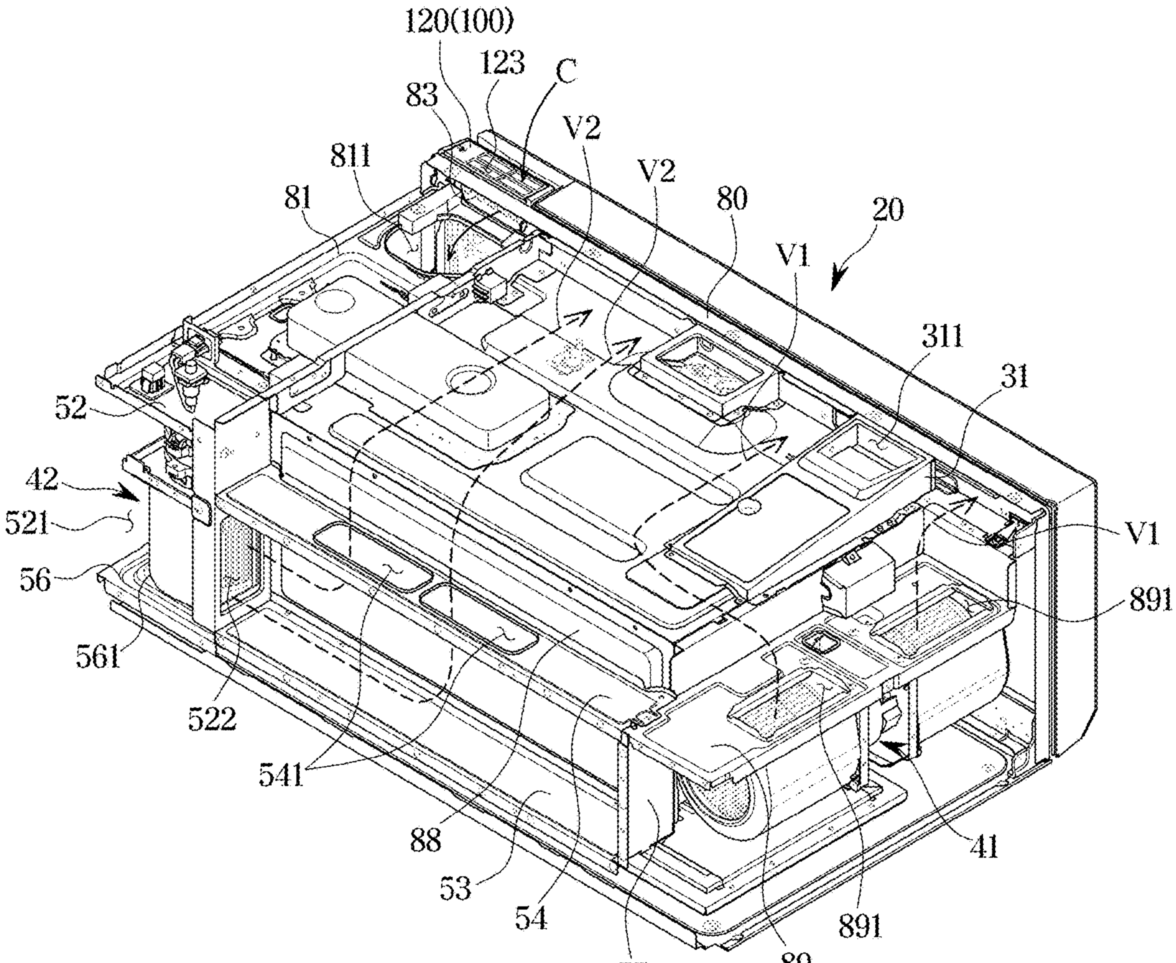

120(100)
123
C
83
V2
811
V2
81
80
20
V1
311
31
52
42
V1
521
56
891
561
522
541
88
41
53
54
891
55
89

FIG. 7

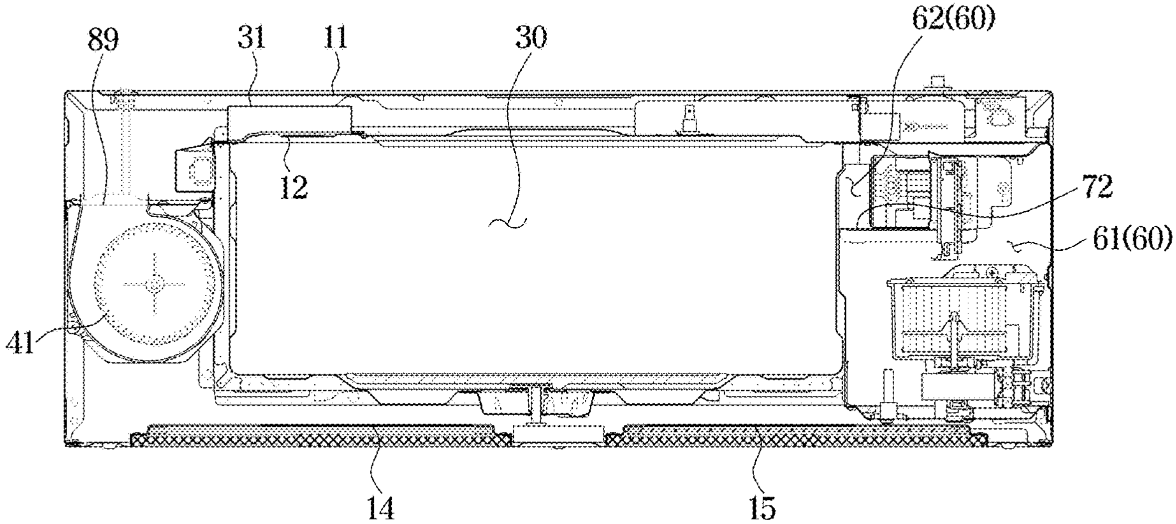


FIG. 8

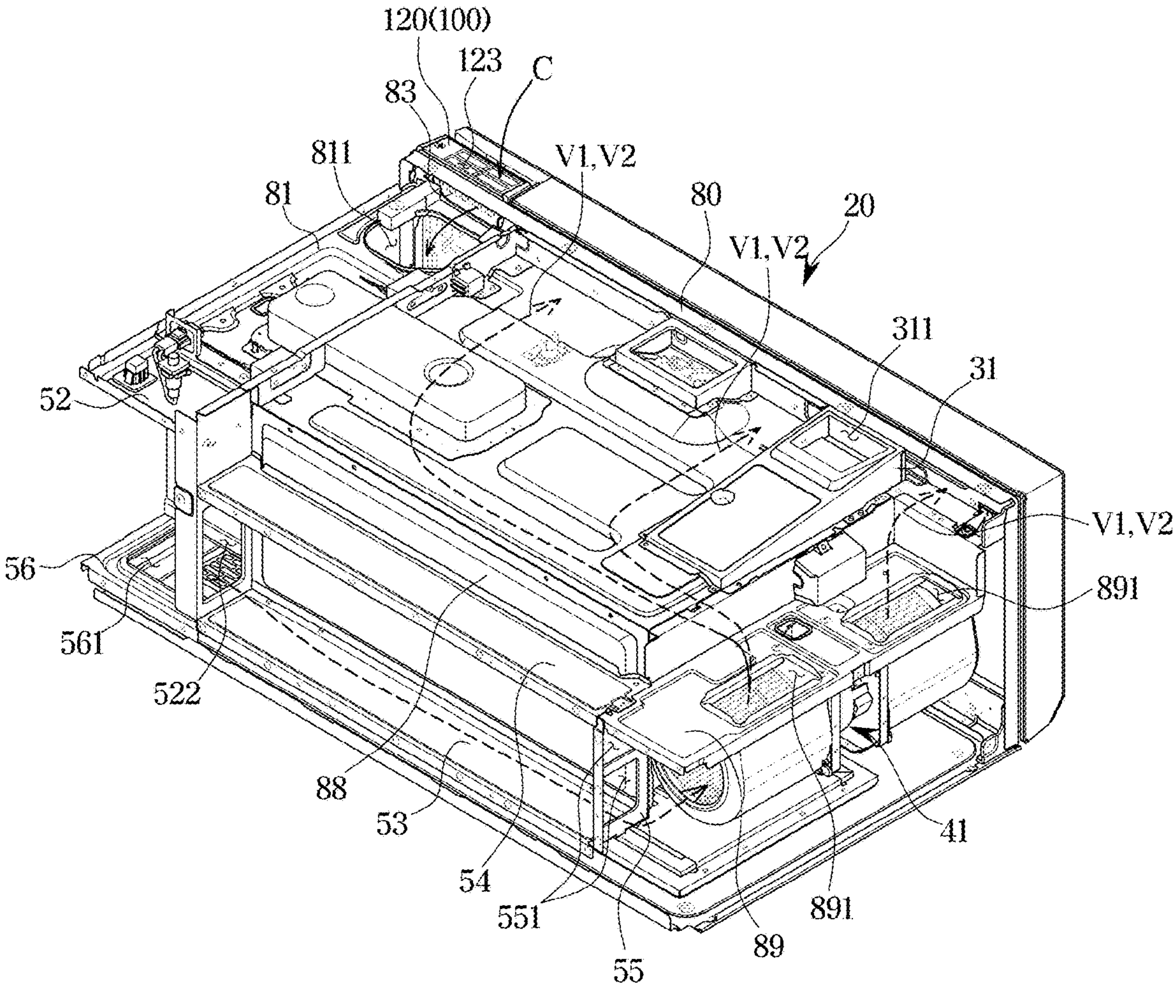

120(100)
123
C
83
811
V1,V2
81
80
20
V1,V2
311
31
52
V1,V2
56
891
561
522
88
53
54
551
55
89
891
41

FIG. 9

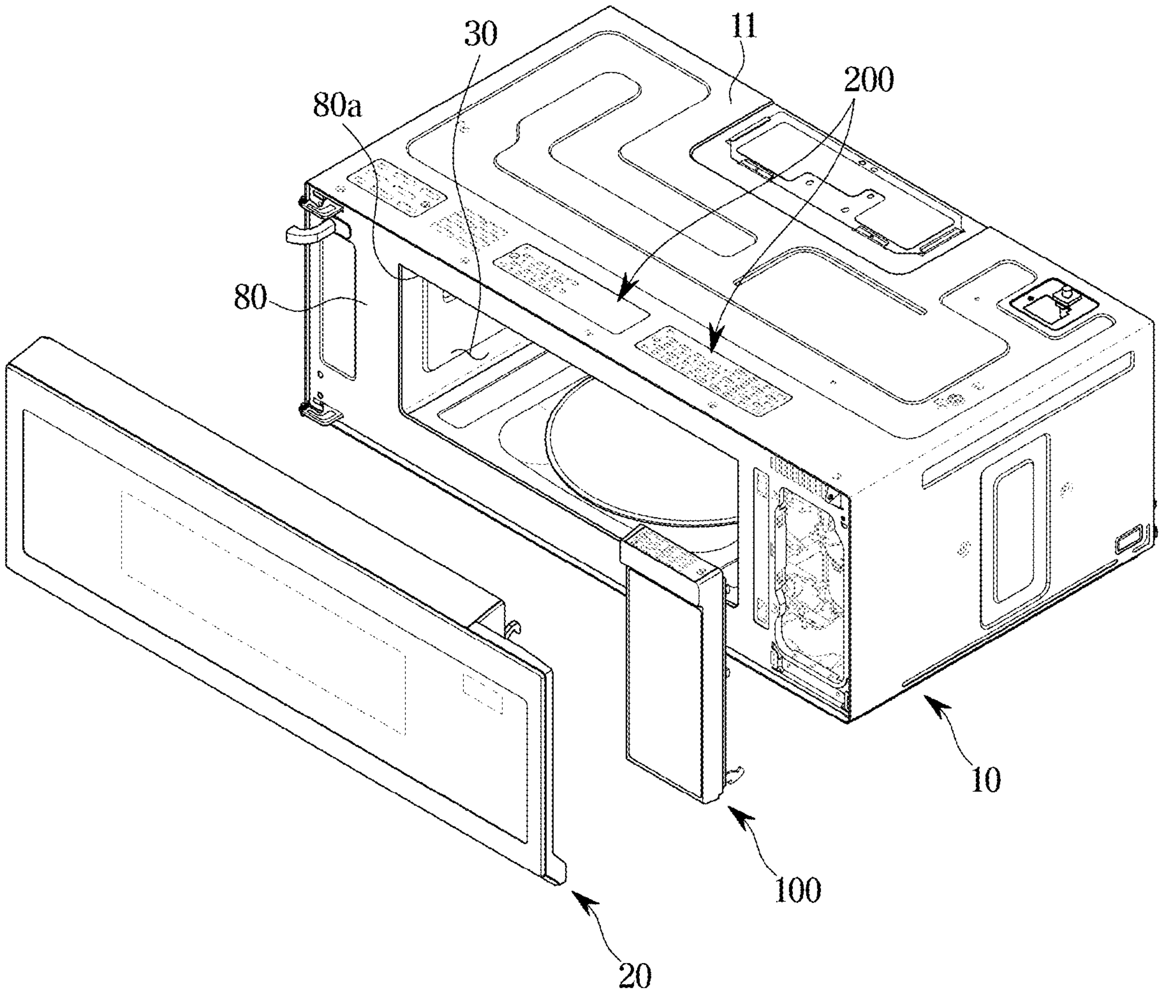


FIG. 10

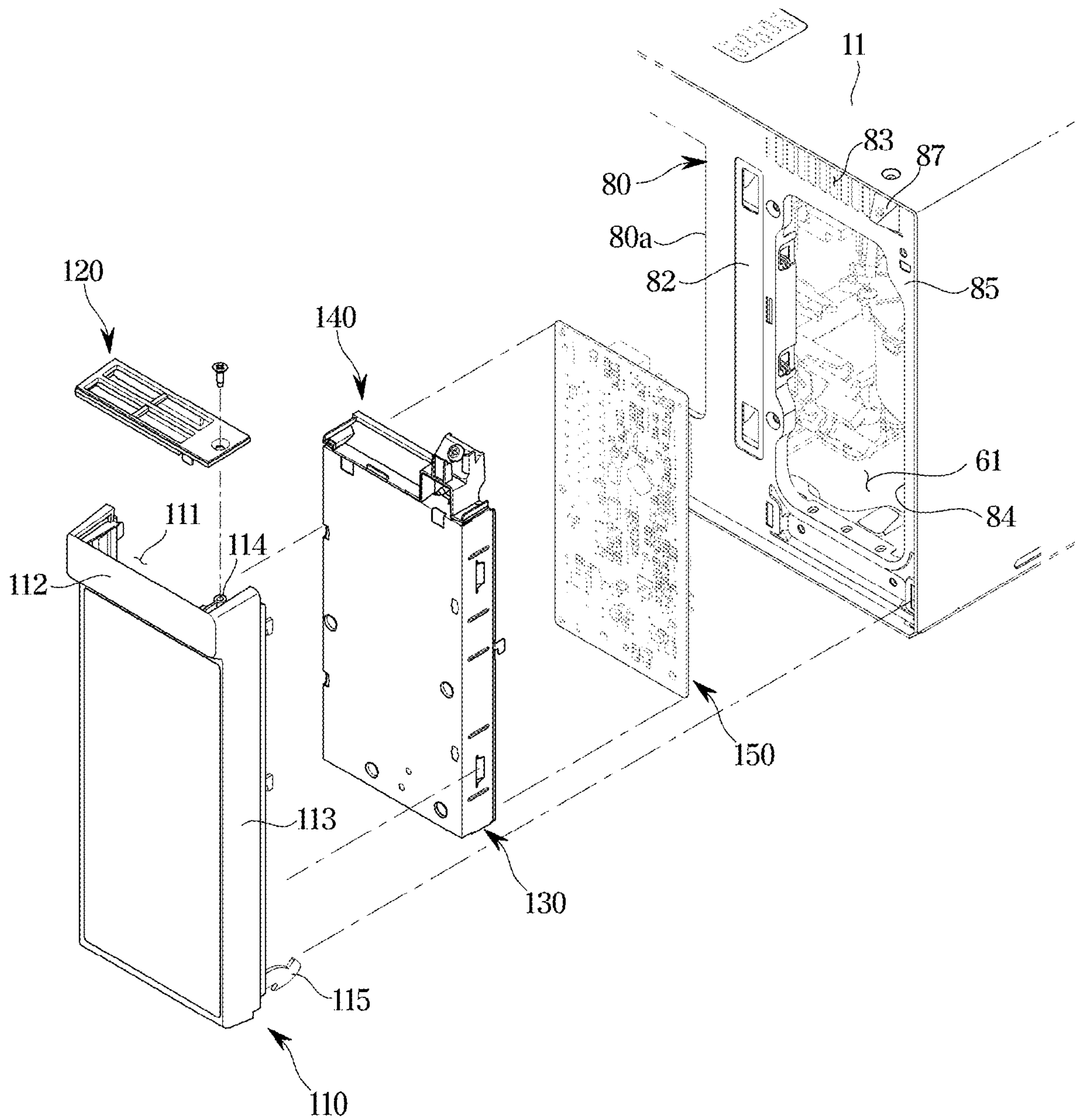


FIG. 11

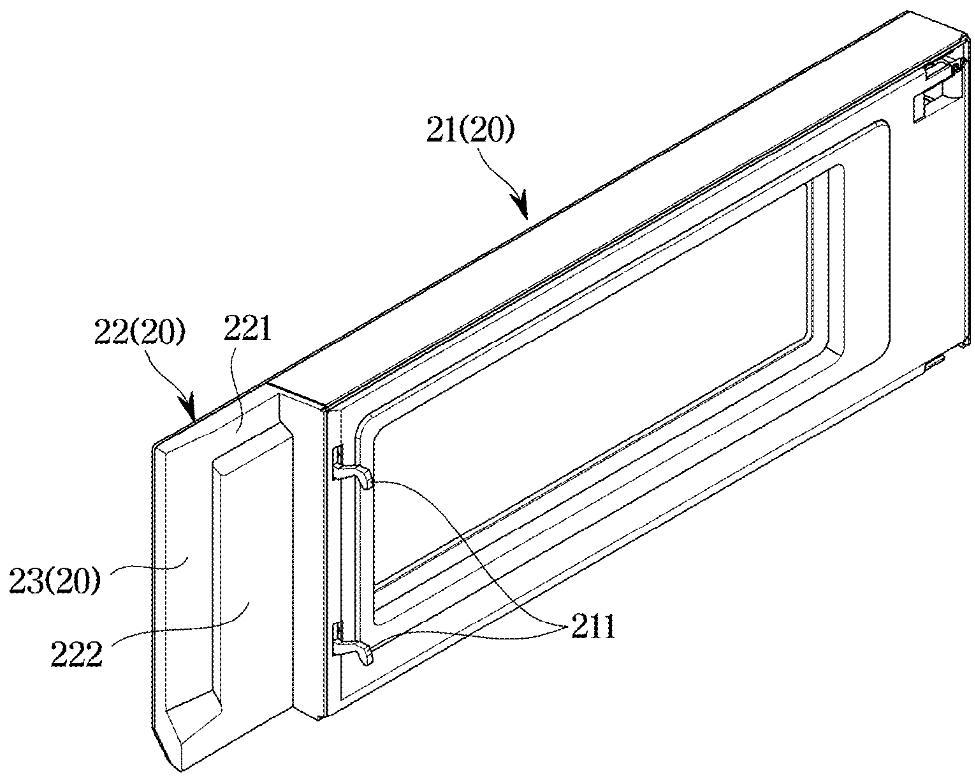


FIG. 12

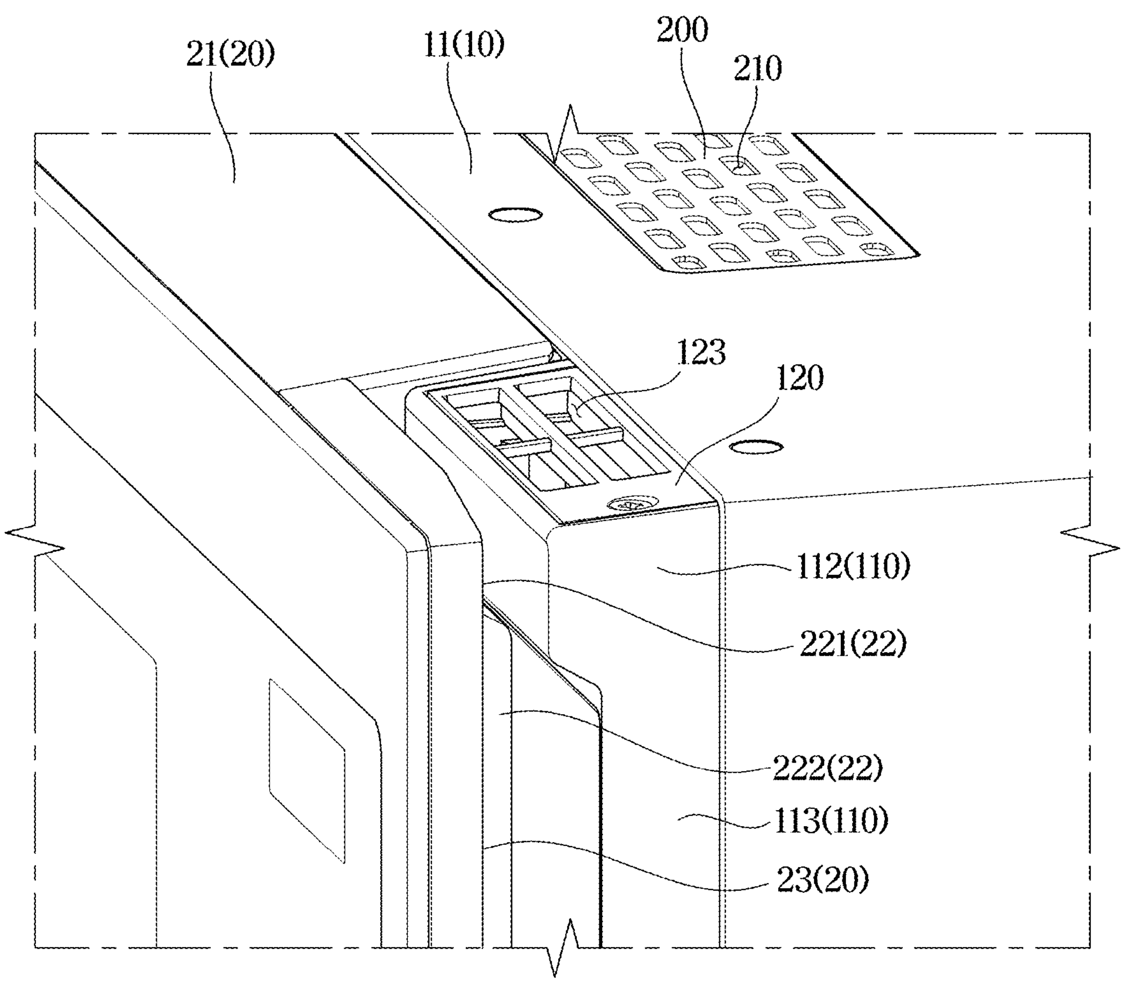


FIG. 13

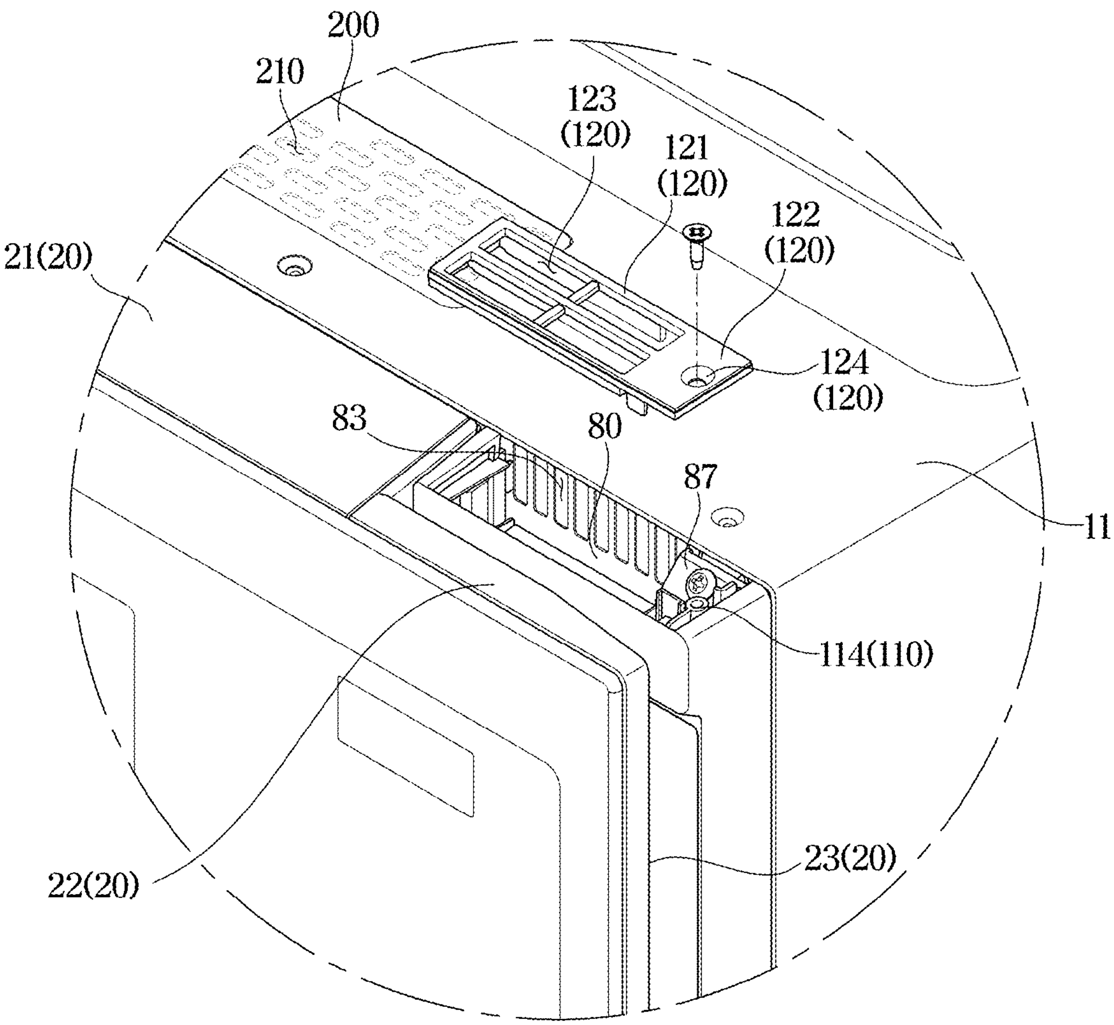

200
210
123
(120)
121
(120)
122
(120)
21(20)
124
(120)
83
80
87
11
114(110)
22(20)
23(20)

FIG. 14

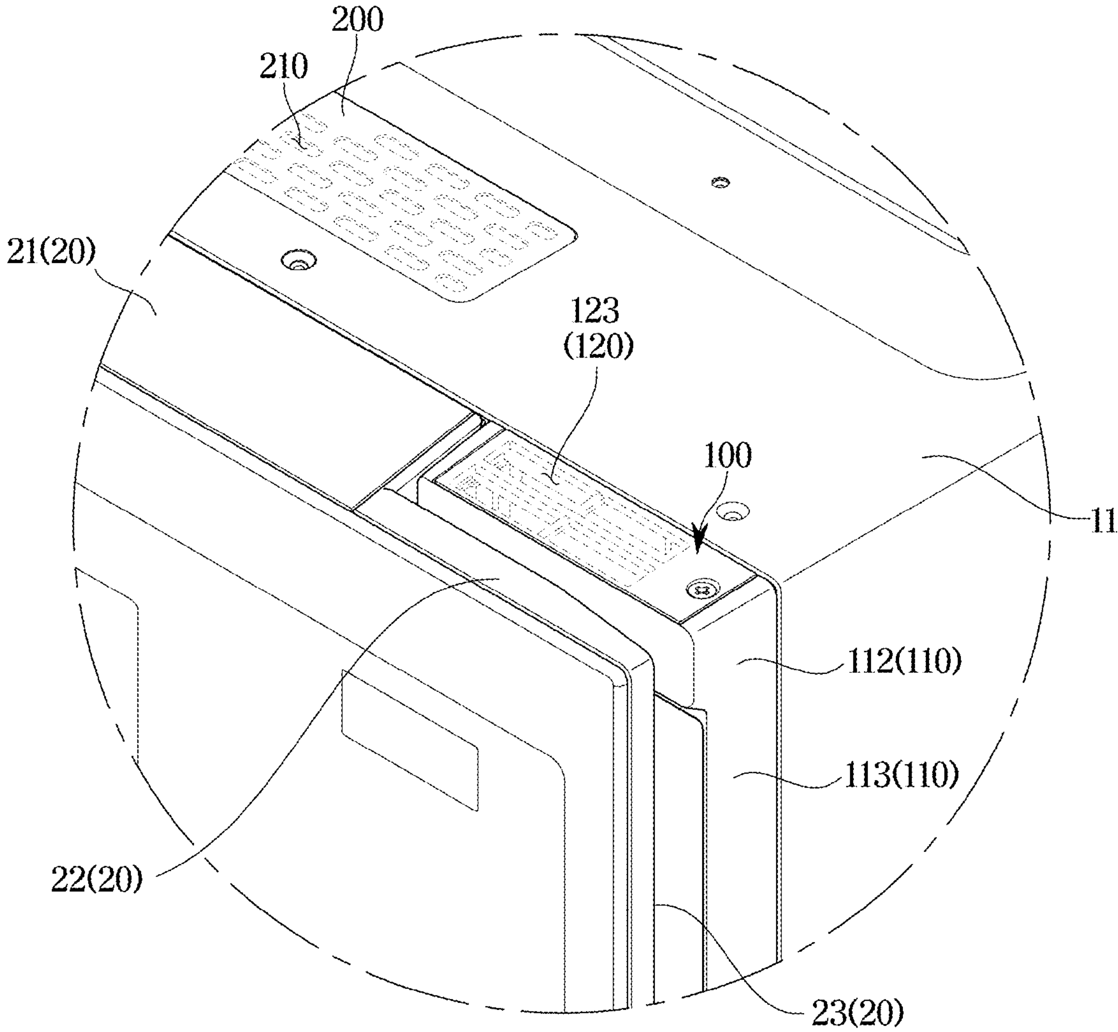


**FIG. 15**

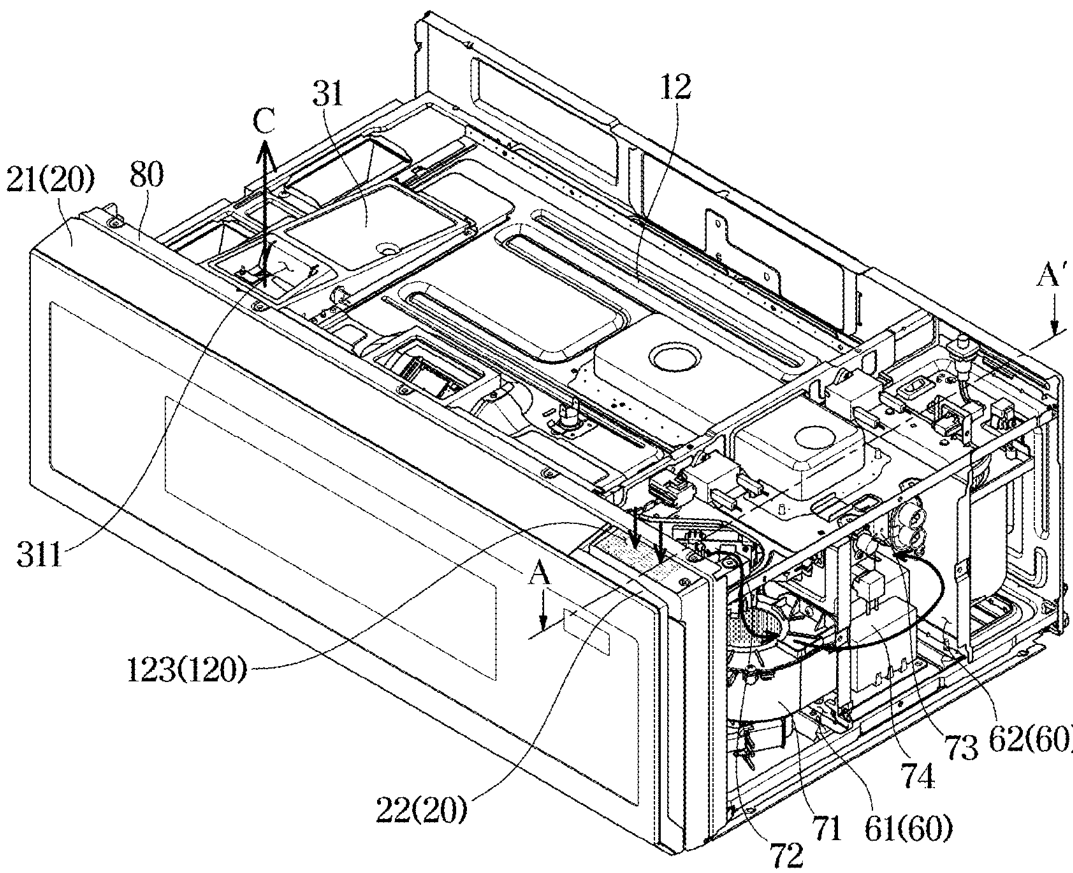


FIG. 16

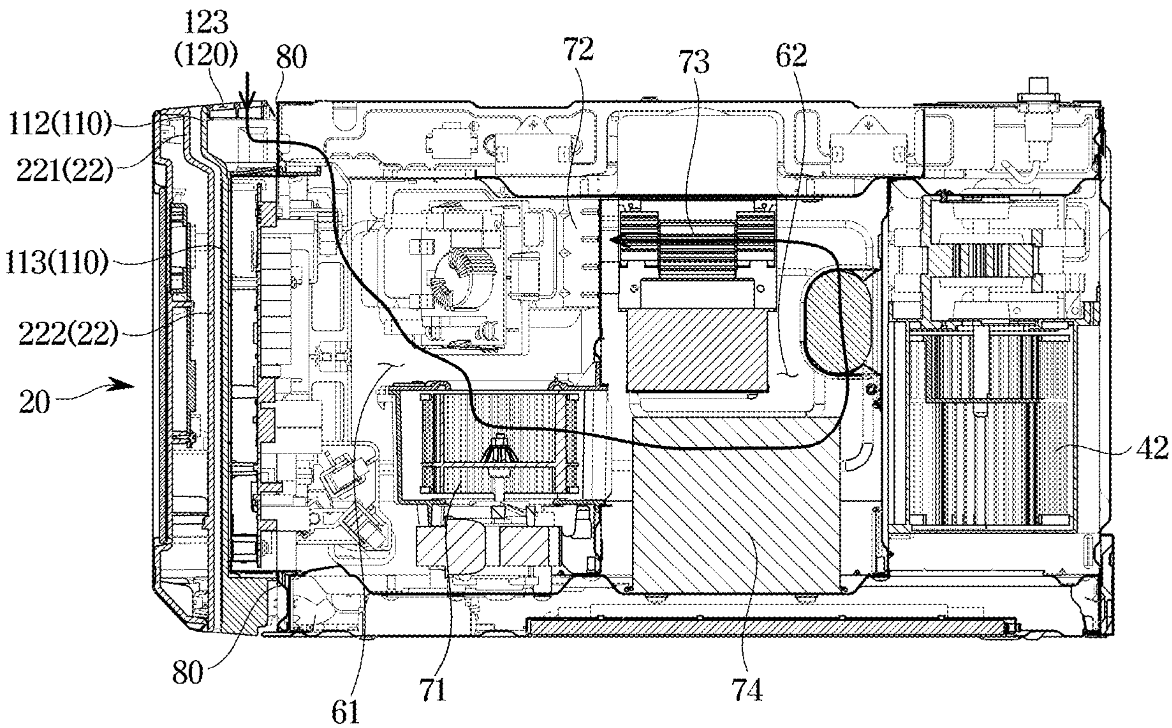


**FIG. 17**

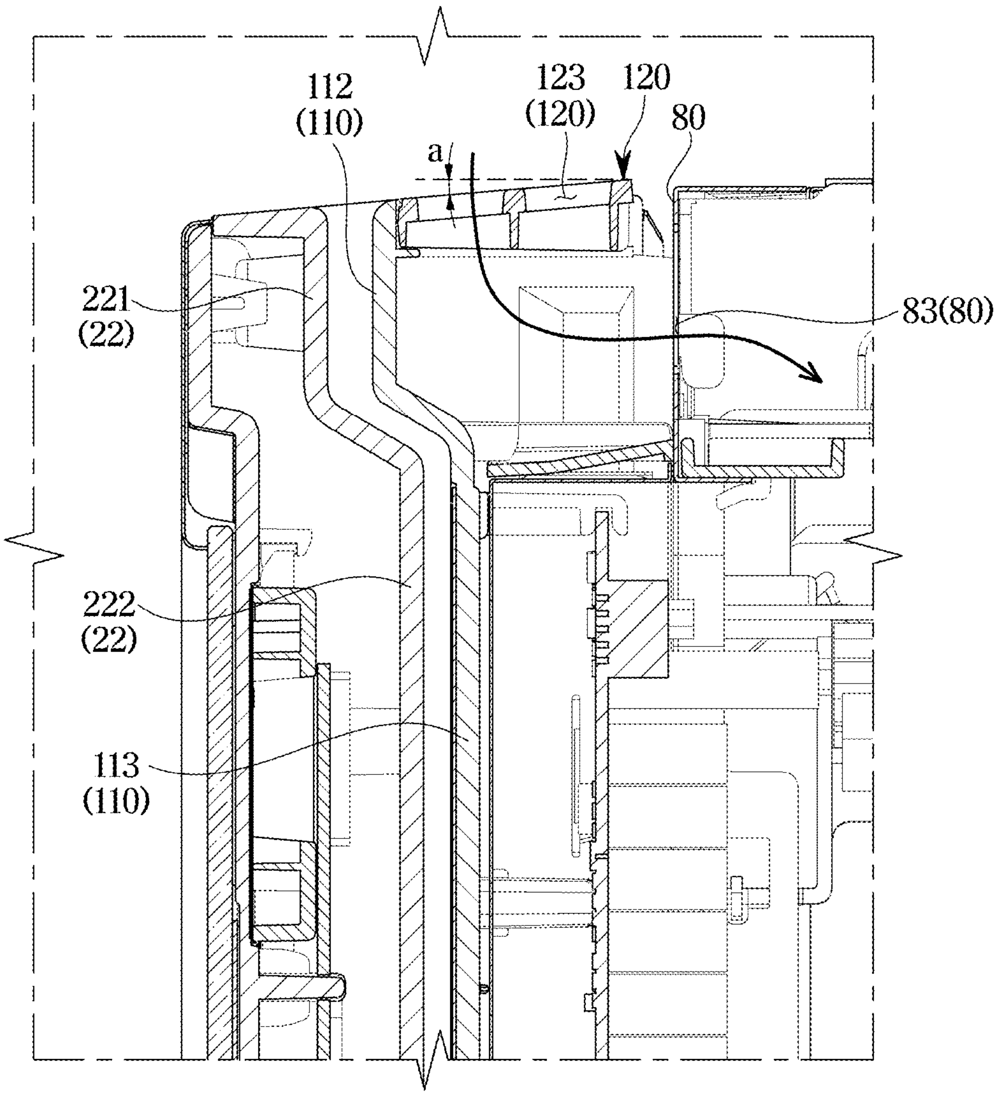
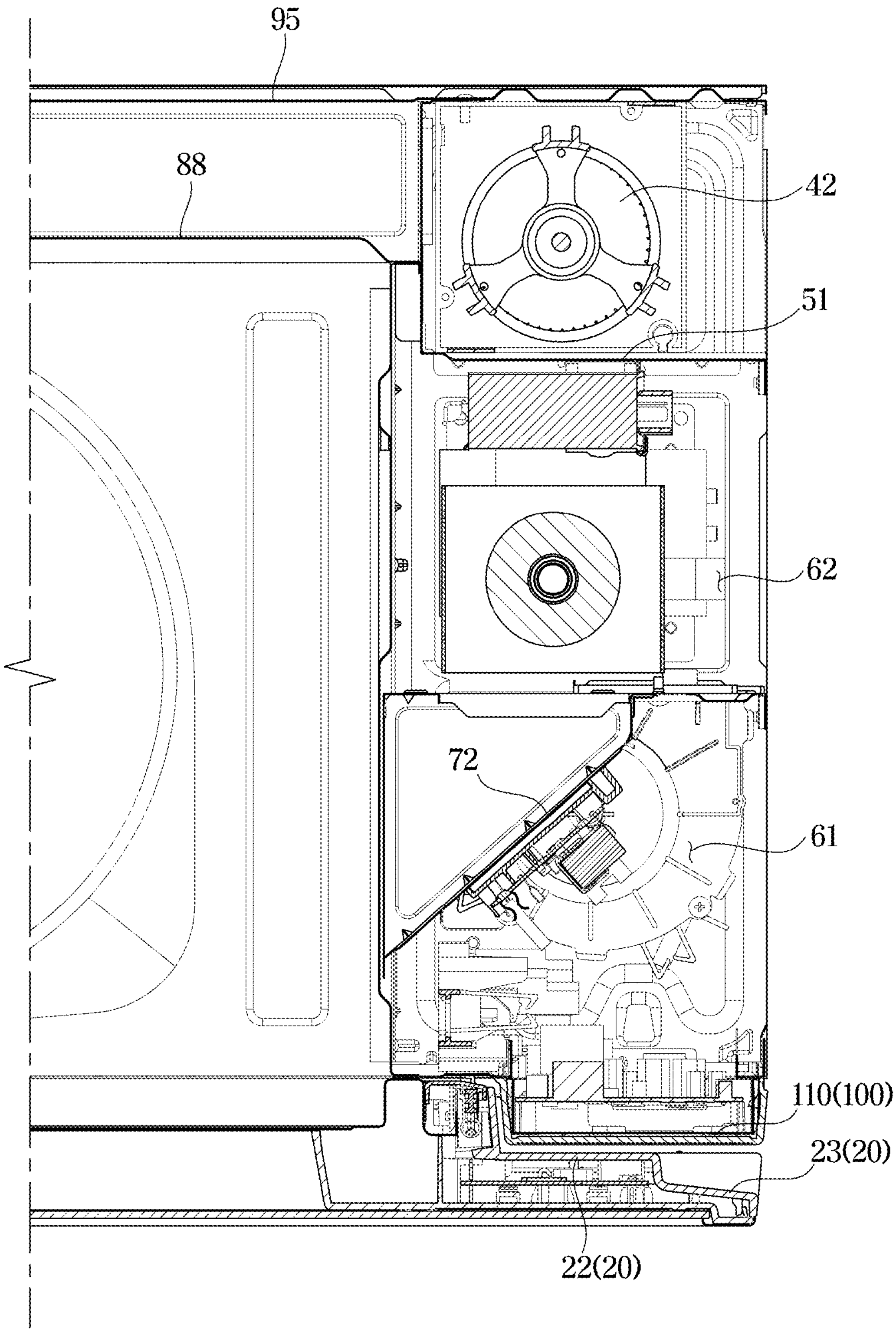


FIG. 18



# COOKING APPARATUS

## CROSS-REFERENCE TO RELATED APPLICATION

**[0001]** This application is a continuation application of U.S. patent application Ser. No. 17/730,357, filed on Apr. 27, 2022 which is a continuation application, under 35 U.S.C. § 111(a), of International Patent Application No. PCT/KR2022/005551, filed on Apr. 18, 2022, which based on and claims benefit of priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0087059, filed on Jul. 2, 2021, and Korean Patent Application No. 10-2021-0096728, filed on Jul. 22, 2021 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

## BACKGROUND

### 1. Field

**[0002]** The disclosure relates to a cooking apparatus with an improved structure, and more particularly, to a cooking apparatus having a cooling flow path with an improved structure.

### 2. Description of the Related Art

**[0003]** A cooking apparatus, which is an apparatus for cooking a cooking material such as food by heating the cooking material, means an apparatus capable of providing various functions related to cooking, such as heating, defrosting, drying, sterilizing, etc. of a cooking material. Examples of the cooking apparatus include an oven, such as a gas oven or en electric oven, a microwave heating apparatus (hereinafter, referred to as a microwave), a gas range, an electric range, an Over The Range (OTR), a gas grill, an electric grill, etc.

**[0004]** The OTR is a microwave combined with a hood function being in charge of ventilation in the kitchen, and the OTR can easily and efficiently cook food in a small installation space. The OTR is positioned above a cooking apparatus, such as a gas range or a cooktop.

**[0005]** The OTR among the cooking apparatuses includes a circulating flow path for sucking polluted air generated below the OTR and discharging the polluted air to the outside of the OTR, and a cooling flow path for cooling a machine room, etc.

**[0006]** However, in the case in which a circulating air outlet through which air flowing along the circulating flow path is discharged is close to a cooling air inlet through which air enters the cooling flow path, the cooling efficiency of the OTR may deteriorate.

## SUMMARY

**[0007]** A cooking apparatus according to a concept of the disclosure includes: a main body forming a cooking room therein and including a front plate with an opening to the cooking room; a case positioned to one side of the opening and formed to accommodate a circuit board therein; a door opening and closing the opening and covering a front portion of the case, wherein a portion of a rear surface of the door is depressed inward to accommodate the case; a circulating air outlet formed at a top of the main body, and configured to filter and discharge air received from below the main body; and a suction grill attachable to and detachable from an upper portion of the case and accommodated in one side of the door, wherein a cooling air inlet, which allows cooling air to enter through the suction grill, is positioned in front of the circulating air outlet.

**[0008]** The cooking apparatus may further include a discharge panel attachable to and detachable from the main body, wherein the circulating air outlet may be formed in the discharge panel, and the discharge panel may be positioned behind the door.

**[0009]** The case may include: a board accommodating portion formed to accommodate the circuit board; and a grill installing portion formed to extend upward from the board accommodating portion and protrude forward from the board accommodating portion.

**[0010]** The suction grill may be coupleable with the grill installing portion, and air may enter the upper portion of the case through the suction grill.

**[0011]** The door may include: a first cover portion depressed along a front direction to cover a front surface of the grill installing portion; a second cover portion to cover a portion of a front surface of the board accommodating portion; and a handle portion positioned to one side of the second cover portion and forming a grip space in front of the board accommodating portion.

**[0012]** The suction grill may be inclined along a front direction of the door.

**[0013]** The cooling air inlet may be a first cooling air inlet, and the front plate may include a second cooling air inlet provided behind the first cooling air inlet to allow air entering the first cooling air inlet to flow to inside of the main body.

**[0014]** The second cooling air inlet may be formed by partially cutting a front portion of the front plate.

**[0015]** The door may include: a sealing portion opening and closing the opening in front of the cooking room; and a cover portion positioned to one side of the sealing portion and to cover the front portion of the case, wherein a thickness of the cover portion may be smaller than a thickness of the sealing portion.

**[0016]** The sealing portion may be positioned in front of the circulating air outlet, and the suction grill may be positioned to the one side of the sealing portion.

**[0017]** The cooling air inlet of the suction grill may be biased with respect to the circulating air outlet.

**[0018]** The cooking apparatus may further include: a machine room provided behind the front plate where air entering through the suction grill enters the machine room; and a guide duct to guide the air entering the machine room to the cooking room.

**[0019]** The cooking apparatus may further include an exhaust duct to discharge air entering the cooking room through the guide duct to outside.

**[0020]** The suction grill may include: a grill portion in which the cooling air inlet is formed; and a coupling portion positioned to one side of the grill portion and coupleable with the case.

**[0021]** The cooking apparatus may further include: an internal housing forming the cooking room; an external housing positioned outside the internal housing and forming an outer appearance of the cooking apparatus; at least one circulating fan positioned between the internal housing and the external housing and configured to suck polluted air from below the main body; and a partition bracket forming a flow path along which air entering the inside of the main body by the at least one circulating fan flows, the partition bracket positioned behind the internal housing and formed to extend horizontally between the internal housing and the external housing.

**[0022]** A cooking apparatus according to another concept of the disclosure includes: a main body forming a cooking room therein; a circulating air inlet provided at a bottom of the main body; a circulating air outlet provided at a top of the main body and discharging air entering the circulating air inlet; a door opening and closing the cooking room; and a control unit coupleable with the main body in such a way as to protrude forward from the main body, wherein a front portion of the control unit is covered by the door, and a cooling air inlet biased with respect to the circulating air outlet is formed at an upper portion of the control unit.

**[0023]** The control unit may include a case, and a suction grill attachable to and detachable from with an upper portion of the case, wherein the cooling air inlet is formed in an upper portion of the suction grill.

**[0024]** The cooling air inlet may be a first cooling air inlet, and the main body may further include a front plate on which the control unit is installed and which is cut to form a second cooling air inlet communicating with the first cooling air inlet.

**[0025]** A cooking apparatus according to another concept of the disclosure includes: a main body including a front plate with an opening and a circulating air outlet provided behind the front plate and configured to filter and discharge air entering inside of the main body; a door including a sealing portion opening and closing the opening and provided in front of the circulating air outlet and a handle portion provided to one side of the sealing portion and depressed inward from a rear surface of the door; and a control unit installed behind the handle portion and formed to protrude toward the door, wherein the control unit includes a case to be accommodated in one side of the door, and a suction grill provided in front of the circulating air outlet, allowing air to enter the one side of the sealing portion, and coupleable with an upper portion of the case, wherein a cooling air inlet is formed in an upper portion of the suction grill.

**[0026]** The circulating air outlet may be provided behind the sealing portion, and the cooling air inlet may be provided to one side of the sealing portion.

**[0027]** Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the disclosure.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0028]** These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:

**[0029]** FIG. **1** is a perspective view of a cooking apparatus according to an embodiment of the disclosure;

**[0030]** FIG. **2** is a bottom perspective view of a cooking apparatus according to an embodiment of the disclosure;

**[0031]** FIG. **3** is a top view of a cooking apparatus according to an embodiment of the disclosure;

**[0032]** FIG. **4** is a side view showing one side of a cooking apparatus according to an embodiment of the disclosure, after an external housing is removed from the cooking apparatus;

**[0033]** FIG. **5** is a side view showing the other side of a cooking apparatus according to an embodiment of the disclosure, after an external housing is removed from the cooking apparatus;

**[0034]** FIG. **6** is a rear perspective view showing a flow of air in a cooking apparatus according to an embodiment of the disclosure after an external housing is removed from the cooking apparatus;

**[0035]** FIG. **7** is a cross-sectional view of a cooking apparatus according to an embodiment of the disclosure;

**[0036]** FIG. **8** is a rear perspective view showing a flow of air in a cooking apparatus according to another embodiment of the disclosure, after an external housing is removed from the cooking apparatus;

**[0037]** FIG. **9** is an exploded perspective view showing a door, a control unit, and a main body in a cooking apparatus according to an embodiment of the disclosure;

**[0038]** FIG. **10** is an enlarged, exploded perspective view showing the control unit and the main body of FIG. **9** according to an embodiment of the disclosure;

**[0039]** FIG. **11** is a rear perspective view of the door shown in FIG. **9** according to an embodiment of the disclosure;

**[0040]** FIG. **12** is an exploded view showing a state in which a control unit of a cooking apparatus according to an embodiment of the disclosure is covered by a door;

**[0041]** FIG. **13** shows a state before a suction grill of a cooking apparatus according to an embodiment of the disclosure is installed in a case;

**[0042]** FIG. **14** shows a state after the suction grill of FIG. **13** is installed in the case according to an embodiment of the disclosure;

**[0043]** FIG. **15** is a perspective view of a cooking apparatus according to an embodiment of the disclosure after an external housing is removed from the cooking apparatus;

**[0044]** FIG. **16** is a cross-sectional view taken along line A-A' of FIG. **15** according to an embodiment of the disclosure;

**[0045]** FIG. **17** is an enlarged view of a upper-front portion of FIG. **16** according to an embodiment of the disclosure; and

**[0046]** FIG. **18** is an enlarged cross-sectional view showing one side of a cooking apparatus according to an embodiment of the disclosure.

## DETAILED DESCRIPTION

**[0047]** Configurations illustrated in the embodiments and the drawings described in the present specification are only the preferred embodiments of the disclosure, and thus it is to be understood that various modified examples, which may replace the embodiments and the drawings described in the present specification, are possible when filing the present application.

**[0048]** Also, like reference numerals or symbols denoted in the drawings of the present specification represent members or components that perform the substantially same functions.

**[0049]** Also, the terms used in the present specification are merely used to describe embodiments, and are not intended to restrict and/or limit the disclosure. It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. It will be understood that when the terms “includes,” “comprises,” “including,” and/or “comprising,” when used in this specification, specify the presence of stated features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

**[0050]** It will be understood that, although the terms including ordinal numbers, such as "first", "second", etc., may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another. For example, a first component could be termed a second component, and, similarly, a second component could be termed a first component, without departing from the scope of the disclosure. As used herein, the term "and/or" includes any and all combinations of one or more of associated listed items.

**[0051]** Throughout the disclosure, the expression "at least one of a, b or c" indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

**[0052]** Therefore, it is an aspect of the disclosure to provide a cooking apparatus having a cooling flow path with improved cooling efficiency.

**[0053]** It is another aspect of the disclosure to provide a cooking apparatus capable of implementing a slim door.

**[0054]** Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

**[0055]** FIG. **1** is a perspective view of a cooking apparatus according to an embodiment of the disclosure. FIG. **2** is a bottom perspective view of a cooking apparatus according to an embodiment of the disclosure. FIG. **3** is a top view of a cooking apparatus according to an embodiment of the disclosure.

**[0056]** A cooking apparatus **1** according to an embodiment of the disclosure may be an Over The Range (OTR). Hereinafter, an OTR will be described as the cooking apparatus **1**.

**[0057]** Referring to FIGS. **1** to **3**, the cooking apparatus **1** may include a main body **10**, and a door **20** coupled with a front portion of the main body **10**.

**[0058]** The cooking apparatus **1** may include a control unit **100** coupled with the front portion of the main body **10**. A front portion of the control unit **100** may be covered by the door **20**. More specifically, the control unit **100** may be coupled with the main body **10** in such a way as to be positioned between the door **20** and the main body **10**.

**[0059]** A portion of a rear surface of the door **20** may be depressed to cover the front portion of the control unit **100**.

**[0060]** In a bottom of the main body **10**, a first circulating air suction portion **14** and a second circulating air suction portion **15** may be formed. More specifically, the first circulating air suction portion **14** and the second circulating air suction portion **15** may be formed in a bottom plate **97** of the main body **10**.

**[0061]** Oil mist, etc. generated below the cooking apparatus **1** may be sucked into inside of the cooking apparatus **1** through the first circulating air suction portion **14** and the second circulating air suction portion **15**. More specifically, the first circulating air suction portion **14** may include a first circulating air inlet, and the second circulating air suction portion **15** may include a second circulating air inlet. The sucked oil mist may be filtered by a filter (not shown) installed inside the cooking apparatus **1** and then discharged to outside of the cooking apparatus **1**. Such a circulating flow path will be described in detail below.

**[0062]** The main body **10** may include a lighting unit **13**. The lighting unit **13** may be positioned in front of the first circulating air suction portion **14** and the second circulating air suction portion **15**. In a case in which a cooktop, etc. is installed below the cooking apparatus **1**, the lighting unit **13** may irradiate light downward from the cooking apparatus **1**.

**[0063]** On a top of the main body **10**, a service panel **91** and a discharge panel **200** may be positioned. More specifically, the service panel **91** may be detachably mounted on a top of an external housing **11**, and the discharge panel **200** may be detachably mounted on the top of the external housing **11**.

**[0064]** Also, a ventilation duct panel **94** may be positioned on the top of the main body **10**. The ventilation duct panel **94** may be detachably mounted on the external housing **11**.

**[0065]** According to an embodiment of the disclosure, two discharge panels **200** may be provided. However, the number of the discharge panels **200** is not limited to two.

**[0066]** The service panel **91** may include a first circulating air outlet **911**. Each of the discharge panels **200** may include a second circulating air outlet **210**.

**[0067]** Polluted air entering the inside of the main body **10** through the first circulating air suction portion **14** and the second circulating air suction portion **15** formed in the bottom of the main body **10** may be filtered and then discharged to the outside of the cooking apparatus **1** through the first circulating air outlet **911** and the second circulating air outlet **210**.

**[0068]** As understood from the above description about the cooking apparatus **1** according to an embodiment of the disclosure, the first circulating air outlet **911** and the second circulating air outlet **210** may be formed in the service panel **91** and the discharge panel **200**, and filtered air may be discharged to indoor through the top of the cooking apparatus **1**, although not limited thereto.

**[0069]** However, in a case in which a separate communicating duct (not shown) connected to outdoor is provided in a space where the cooking apparatus **1** is installed, the service panel **91** may not include the first circulating air outlet **911**, and the discharge panel **200** may also not include the second circulating air outlet **210**.

**[0070]** Instead, a circulating air outlet may be formed in the ventilation duct panel **94**. In this case, oil mist entered the first circulating air inlet and the second circulating air inlet may be filtered, flow to the communicating duct through the ventilation duct panel **94** positioned in a rear portion of the top of the cooking apparatus **1**, and then be discharged to the outdoor.

**[0071]** In other words, because the first circulating air outlet **911** is not formed in the service panel **91**, the second circulating air outlet **210** is not formed in the discharge panel **200**, and the circulating air outlet is formed in the ventilation duct panel **94**, all filtered oil mist may be discharged to the outdoor without being discharged to the indoor.

**[0072]** Therefore, indoor air may be kept cleaner and temperature of cooling air entering through a cooling flow path C which will be described below may be lowered, which may result in an increase of cooling efficiency of the cooking apparatus **1**.

**[0073]** Also, because the service panel **91**, the discharge panel **200**, and the ventilation duct panel **94** are detachable from the external housing **11**, the service panel **91**, the discharge panel **200**, and the ventilation duct panel **94** may be installed in the external housing **11** as necessary.

**[0074]** More specifically, in a case in which filtered air needs to be discharged to the indoor because no separate communicating duct (not shown) is provided, the service panel **91** and the discharge panel **200** having circulating air outlets may be installed in the external housing **11**, and the ventilation duct panel **94** having no circulating air outlet may be installed in the external housing **11**.

**[0075]** In contrast, in a case in which filtered air does not need to be discharged to the indoor because a separate communicating duct (not shown) is provided, the service panel **91** and the discharge panel **200** having no circulating air outlets may be installed in the external housing **11**, and the ventilation duct panel **94** having a circulating air outlet may be installed in the external housing **11**.

**[0076]** The control unit **100** may be coupled with the front portion of the main body **10**, and a suction grill **120** may be coupled with an upper portion of the control unit **100**.

**[0077]** The suction grill **120** may be positioned in front of the discharge panel **200**. More specifically, the suction grill **120** may be accommodated in the depressed portion of the door **20** and aligned with the door **20** at one side of the door **20**. Also, the suction grill **120** may be positioned behind a portion of the door **20**.

**[0078]** Outside air of the cooking apparatus **1** may enter the inside of the main body **10** through the suction grill **120**. The suction grill **120** may include a first cooling air inlet **123**. Accordingly, air may enter the inside of the main body **10** through the first cooling air inlet **123** formed in the suction grill **120** to cool a machine room **60** (see FIG. **4**).

**[0079]** Air entering through the suction grill **120** may cool the machine room **60**, enter a cooking room **30** (see FIG. **7**) to cool the rear surface of the door **20**, and then be discharged through a cooling air outlet **93** formed in the top of the main body **10**. More specifically, the cooling air outlet **93** may be formed by cutting a portion of the top of the external housing **11**.

**[0080]** FIG. **4** is a side view showing one side of a cooking apparatus according to an embodiment of the disclosure, after an external housing is removed from the cooking apparatus. FIG. **5** is a side view showing the other side of a cooking apparatus according to an embodiment of the disclosure, after an external housing is removed from the cooking apparatus.

**[0081]** Referring to FIG. **4**, the cooking apparatus **1** may include the machine room **60**. More specifically, the machine room **60** may include a first machine room **61** and a second machine room **62**.

**[0082]** An internal housing **12** (see FIG. **7**) of the main body **10** may include a top plate **81** and a front plate **80**.

**[0083]** The control unit **100** may be installed in front of the front plate **80**. The first machine room **61** and the second machine room **62** may be provided behind the front plate **80**.

**[0084]** The machine room **60** may be partitioned inside the main body **10** by the front plate **80**, the top plate **81**, a cooling divider **51** (see FIG. **18**), and a circulating plate **56**. Also, the machine room **60** may be partitioned into two spaces by a partition plate **63** that partitions the first machine room **61** from the second machine room **62**. The cooling divider **51** may be provided as one surface of a rear bracket **52**.

**[0085]** The circulating plate **56** may form a bottom of the machine room **60**. The circulating plate **56** may be spaced from a bottom plate **97** of the external housing **11**.

**[0086]** The cooling divider **51** may be positioned above the circulating plate **56** to partition the machine room **60**.

**[0087]** As a result of partitioning of the machine room **60** by the circulating plate **56** and the cooling divider **51**, oil mist entered through the second circulating air suction portion **15** formed in the bottom of the main body **10** may be prevented from entering the machine room **60**. Thereby, components installed inside the machine room **60** may be protected from oil mist or other foreign materials.

**[0088]** A cooling fan **71** may be accommodated in the first machine room **61**. The cooling fan **71** may form a flow of air to cause outside air to enter the suction grill **120**.

**[0089]** Also, the control unit **100** may be mounted on the front plate **80** in such a way as to communicate with the first machine room **61**, which will be described below, and accordingly, air entering the suction grill **120** may cool the control unit **100**. Details about this will be described below.

**[0090]** A magnetron **73** and a transformer **74** may be accommodated in the second machine room **62**. The magnetron **73** and the transformer **74** may generate a high frequency. The generated high frequency may be supplied to the cooking room **30** to heat a material to be cooked.

**[0091]** Hereinafter, the cooling flow path C of the cooking apparatus **1** will be described.

**[0092]** A portion of the partition plate **63** partitioning the first machine room **61** from the second machine room **62** may open. The partition plate **63** may include a lower opening (not shown) communicating with the cooling fan **71** such that air sucked into the suction grill **120** by the cooling fan **71** passes through the first machine room **61** and enters the second machine room **62**. Also, the partition plate **63** may include an upper opening (not shown) communicating the second machine room **62** with the guide duct **72**.

**[0093]** Air entering the second machine room **62** may cool the transformer **74** and the magnetron **73** and then flow to the guide duct **72** installed in the partition plate **63**. The guide duct **72** may be positioned in the first machine room **61** to form a space through which the second machine room **62** communicates with the cooking room **30**.

**[0094]** Accordingly, air entering the guide duct **72** may flow to the cooking room **30** and cool the rear surface of the door **20** during cooking. Thereafter, the air may flow to an exhaust duct **31** communicating with the cooking room **30** at an upper surface of the cooking room **30** and be discharged to the outside of the cooking apparatus **1** through the cool air outlet **93** communicating with the exhaust duct **31**.

**[0095]** The cooking apparatus **1** may include circulating flow paths V1 and V2 for sucking oil mist and filtering the oil mist, in addition to the cooling flow path C for cooling the machine room **60** and the door **20**.

**[0096]** The cooking apparatus **1** may include a second circulating fan **42** and the rear bracket **52**. The rear bracket **52** may include a circulating fan accommodating portion **521** accommodating the second circulating fan **42**.

**[0097]** A front portion of the circulating plate **56** may be closed to partition the machine room **60**. However, the circulating plate **56** may include a communicating portion **522** opening to communicate with the circulating fan accommodating portion **521** of the rear bracket **52**, at a rear portion. Because the communicating portion **522** is formed in the circulating plate **56**, oil mist may enter the second circulating air suction portion **15** by a suction force generated by the second circulating fan **42** and be filtered.

**[0098]** Referring to FIG. **5**, the cooking apparatus **1** may include a first circulating fan **41**. The first circulating fan **41** may generate a suction force to cause air to enter the first circulating air suction portion **14** formed in the bottom plate **97**. A side plate **89** of the internal housing **12** may be installed above the first circulating fan **41**. A portion of the side plate **89** may be cut such that received air flows to an upper portion of the first circulating fan **41**.

**[0099]** Accordingly, the cooking apparatus **1** according to an embodiment of the disclosure may include the first circulating fan **41** and the second circulating fan **42** at both sides. Accordingly, inside the cooking apparatus **1**, a plurality of circulating flow paths V**1** and V**2** having a first circulating flow path V**1** through which air flows by the first circulating fan **41** and a second circulating flow path V**2** through which air flows by the second circulating fan **42** may be formed. Details about this will be described below.

**[0100]** FIG. **6** is a rear perspective view showing a flow of air in a cooking apparatus according to an embodiment of the disclosure after an external housing is removed from the cooking apparatus. FIG. **7** is a cross-sectional view of a cooking apparatus according to an embodiment of the disclosure.

**[0101]** Hereinafter, the first circulating flow path V**1**, the second circulating flow path V**2**, and the cooling flow path C, according to an embodiment of the disclosure, will be described.

**[0102]** Referring to FIGS. **5** and **6**, air may enter the inside of the main body **10** through the first circulating air suction portion **14** formed in the bottom of the main body **10**. The air may be polluted air containing oil mist.

**[0103]** The air may be filtered and pass through the first circulating fan **41**. The air passed through the first circulating fan **41** may flow to the side plate **89** positioned above the first circulating fan **41**.

**[0104]** The side plate **89** may include a first communicating hole **891**. Air passed through the first circulating fan **41** through the first communication hole **891** may flow over the internal housing **12**.

**[0105]** The flowing air may be discharged to the outside of the cooking apparatus **1** through the first circulating air outlet **911** and the second circulating air outlet **210** formed in a front portion of the top of the main body **10**, as shown in FIG. **3**. This is referred to as the first circulating flow path V**1**.

**[0106]** However, in a case in which a separate communicating duct is provided to discharge filtered oil mist to the outdoor, as described above, air entering the first circulating air suction portion **14** may be discharged to the outside of the cooking apparatus **1** through the circulating air outlet formed in the ventilation duct panel **94**.

**[0107]** Referring to FIGS. **4** and **6**, air may enter the inside of the main body **10** through the second circulating air suction portion **15** formed in the bottom of the main body **10**. The air may also be polluted air containing oil mist.

**[0108]** The air may be filtered and pass through the second circulating fan **42**. More specifically, the air may enter the circulating fan accommodating portion **521** of the rear bracket **52** through a through hole **561** formed in the circulating plate **56**.

**[0109]** The rear bracket **52** may include the communicating portion **522**. The communicating portion **522** may be formed between a back plate **95** of the external housing **11** and a rear plate **88** of the internal housing **12**. Accordingly, air entering the through hole **561** may flow behind the internal housing **12** through the communicating portion **522**.

**[0110]** The cooking apparatus **1** may include a bottom bracket **53**, a partition bracket **54**, and a circulating divider **55**.

**[0111]** The bottom bracket **53** and the partition bracket **54** may be positioned between the external housing **11** and the internal housing **12**. More specifically, the bottom bracket **53** and the partition bracket **54** may be positioned behind the internal housing **12** and extend horizontally. The bottom bracket **53** may be positioned below the partition bracket **54** to face the partition bracket **54**.

**[0112]** The circulating bracket **55** may be positioned between the external housing and the internal housing **12**. More specifically, the circulating divider **55** may be positioned behind the internal housing and extend vertically. That is, the circulating divider **55** may face the communicating portion **522** of the rear bracket **52**.

**[0113]** The partition bracket **54** may include a second communicating hole **541**. The second communicating hole **541** may be formed by cutting a portion of the partition bracket **54**. Air may pass through the second communicating hole **541**.

**[0114]** Accordingly, the rear bracket **52**, the bottom bracket **53**, the partition bracket **54**, and the circulating divider **55** may form a portion of a flow path along which air entering the second circulating air suction portion **15** flows.

**[0115]** The air entering the second circulating air suction portion **15** may pass through the through hole **561** of the circulating plate **56**, pass through the second circulating fan **42**, pass through the communicating portion **522** of the rear bracket **52**, and then flow between the bottom bracket **53** and the partition bracket **54**.

**[0116]** The circulating divider **55** may be closed without any opening and positioned between the partition bracket **54** and the bottom bracket **53**. Accordingly, the air may flow over the internal housing **12** from behind the internal housing **12** through the second communicating hole **541** of the partition bracket **54**.

**[0117]** The air moved to above the internal housing **12** through the second communicating hole **541** may be again discharged to the outside of the cooking apparatus **1** through the first circulating air outlet **911** and the second circulating air outlet **210** formed in the front top of the main body **10**. This is referred to as the second circulating flow path V**2**.

**[0118]** However, in a case in which a separate communicating duct is provided to discharge filtered oil mist to the outdoor, as described above, air entering the second circulating air suction portion **15** may be discharged to the outside of the cooking apparatus **1** through the circulating air outlet formed in the ventilation duct panel **94**.

**[0119]** Because the circulating divider **55** is closed, air entering by the first circulating fan **41** and air entering by the second circulating fan **42** may flow separately below the internal housing **12** and then be mixed above the internal housing **12**.

**[0120]** Referring to FIGS. **6** and **7**, air entering the suction grill **120** positioned on the upper portion of the control unit **100** may cool various electronic components by flowing through the first machine room **61** and the second machine room **62**. More specifically, air entering the first cooling air inlet **123** of the suction grill **120** may enter the inside of the main body **10** through a second cooling air inlet **83** of the front plate **80**. Thereafter, the air may enter the inside of the machine room **60** through an incision portion **811** formed in the top plate **81** of the internal housing **12**.

**[0121]** Air entering the first machine room **61** may enter the second machine room **62** through the cooling fan **71**, and the air entering the second machine room **62** may be guided to the cooking room **30** by the guide duct **72** communicating with the cooking room **30** at the side surface of the cooking room **30**. The air guided to the cooking room **30** may cool the rear surface of the door **20**, and then flow to the exhaust duct **31** communicating with the cooking room **30** at the upper surface of the cooking room **30**. The exhaust duct **31** may communicate with the cooling air outlet **93**. Accordingly, completely cooled air may be discharged to the outside of the cooking apparatus **1** through the cooling air outlet **93**. This is referred to as the cooling flow path C.

**[0122]** FIG. **8** is a rear perspective view showing a flow of air in a cooking apparatus according to another embodiment of the disclosure, after an external housing is removed from the cooking apparatus.

**[0123]** A cooking apparatus according to another embodiment of the disclosure may include a single circulating fan **41**, unlike the cooking apparatus **1** according to an embodiment of the disclosure. More specifically, the cooking apparatus may include the first circulating fan **41**.

**[0124]** Accordingly, the cooking apparatus according to another embodiment of the disclosure may form a single circulating flow path, unlike the cooking apparatus **1** according to an embodiment of the disclosure.

**[0125]** Also, the cooking apparatus according to another embodiment of the disclosure may include the same configurations as the cooking apparatus **1** according to an embodiment of the disclosure, except for different configurations of the circulating divider **55** and the partition bracket **54**.

**[0126]** Accordingly, descriptions about the same configurations included in the cooking apparatus **1** as described above will be omitted, and the same configurations will be assigned the same reference numerals.

**[0127]** The cooking apparatus according to another embodiment of the disclosure may include the first circulating fan **41** positioned between the internal housing **12** and the external housing **11**. The first circulating fan **41** may cause air to be sucked into the first circulating air inlet and the second circulating air inlet.

**[0128]** More specifically, air entering the first circulating air inlet may pass through the first circulating fan **41** and pass through the first communicating hole **891** of the side plate **89**. Air passed through the first communicating hole **891** may be discharged to the outside of the cooking apparatus through the first circulating air outlet **911** of the service panel **91** and a second circulating air outlet **921** of a discharge panel **92**.

**[0129]** However, in a case in which a separate communicating duct (not shown) is provided to discharge filtered oil mist to the outdoor, as described above, air entering the first circulating air inlet may be discharged to the outside of the cooking apparatus through the circulating air outlet formed in the ventilation duct panel **94**.

**[0130]** This may be the same as the first circulating flow path V1 of the cooking apparatus **1** according to an embodiment of the disclosure.

**[0131]** However, air entering the second circulating air inlet may pass through the through hole **561** of the circulating plate **56** and flow to a space between the partition bracket **54** and the bottom bracket **53** through the communicating portion **522** of the rear bracket **52**.

**[0132]** The partition bracket **54** may be closed without any opening. More specifically, no second communicating hole may be formed in the partition bracket **54**, unlike the cooking apparatus **1** according to an embodiment of the disclosure.

**[0133]** However, the partition bracket **54** may be closed, and a second communication hole **551** may be formed in the circulating divider **55**. Accordingly, air entering the second circulating air inlet may enter the space between the partition bracket **54** and the bottom bracket **53** and flow toward the first circulating fan **41** through the second communicating hole **551** of the circulating divider **55**.

**[0134]** Thereafter, air passed through the first circulating fan **41** may flow to the upper portion of the internal housing **12** through the first communicating hole **891** of the side plate **89**. The air may be discharged to the outside of the cooking apparatus through the first circulating air outlet **911** of the service panel **91** and the second circulating air outlet **921** of the discharge panel **92**.

**[0135]** However, in a case in which a separate communicating duct (not shown) is provided to discharge filtered oil mist to the outdoor, as described above, air entering the second circulating air inlet may be discharged to the outside of the cooking apparatus through the circulating air outlet formed in the ventilation duct panel **94**.

**[0136]** Accordingly, the cooking apparatus according to another embodiment of the disclosure may suck air into the first circulating air inlet and the second circulating air inlet with a suction force of the first circulating fan **41** by closing the partition bracket **54** and opening the circulating divider **55**, unlike the cooking apparatus **1** according to an embodiment of the disclosure. That is, the first circulating flow path V**1** and the second circulating flow path V**2** may be formed as a single flow path via the first circulating fan **41**.

**[0137]** Accordingly, a structure of the cooking apparatus may be more simplified, which reduces manufacturing cost. Also, because it is possible to replace configurations of the partition bracket **54** and the circulating divider **55** according to a user's need, complicating an internal design according to the number of circulating fans may be not needed.

**[0138]** FIG. **9** is an exploded perspective view showing a door, a control unit, and a main body in a cooking apparatus according to an embodiment of the disclosure. FIG. **10** is an enlarged, exploded perspective view showing the control unit and the main body of FIG. **9**.

**[0139]** Referring to FIGS. **9** and **10**, the control unit **100** may be installed on the front portion of the main body **10**. The door **20** may be rotatably coupled with the main body **10** to cover the front portions of the main body **10** and the control unit **100**.

**[0140]** More specifically, the main body **10** may include the front plate **80**. The front plate **80** may be installed on the front portion of the external housing **11**. The discharge panel **200** and the service panel **91** may be detachably coupled with the top of the external housing **11**. Also, in the top of the external housing **11**, the cooling air outlet **93** may be incised and formed, although not limited thereto. However, the cooling air outlet **93** may also be formed in a separate panel that is detachable from the external housing **11**.

**[0141]** The front plate **80** may include an opening **80***a* communicating with the cooking room **30**, and an installing portion **85** which is positioned to one side of the opening **80***a* and on which the control unit **100** is installed.

**[0142]** Also, the front plate **80** may include a door fixing portion **82** positioned between the opening **80***a* and the installing portion **85**. More specifically, a door latch **211** which will be described below may be inserted into the door fixing portion **82** to maintain a state in which the door **20** closes the cooking room **30**.

**[0143]** The service panel **91** in which the first circulating air outlet **911** is formed and the discharge panel **200** in which the second circulating air outlet **210** is formed may be biased to one side of the main body **10**. More specifically, the service panel **91** and the discharge panel **200** may be arranged to be biased in a direction being maximally away from the control unit **100**. This will be described in detail, below.

**[0144]** A portion of the rear surface of the door **20** may be depressed inward to accommodate the control unit **100**. More specifically, the door **20** may be rotatably coupled with the main body **10** in such a way as to cover the front portion of the control unit **100**. A user may open or close the door **20** by using a handle portion **23** formed between the control unit **100** and the door **20**. Details about the door **20** will be described in detail, below.

**[0145]** The control unit **100** may include a case **110** and the suction grill **120** installed on an upper portion of the case **110**.

**[0146]** The case **110** and the suction grill **120** may form an outer appearance of the control unit **100**. More specifically, an opening **111** may be formed in the upper portion of the case **110**, and the suction grill **120** may be coupled with the upper portion of the case **110** to cover the opening **111** of the case **110**.

**[0147]** Also, the case **110** may be in a shape of a box of which the rear portion opens.

**[0148]** The case **110** may be installed on the one side of the front plate **80** and accommodate a circuit board **150** therein.

**[0149]** The case **110** may include an insertion lag **115** inserted into an insertion hole of the installing portion **85** of the front plate **80**. The insertion lag **115** may extend from a rear portion of the case **110** toward the front plate **80**. The insertion lag **115** may be substantially in a shape of a hook and fix the case **110** on the front plate **80**.

**[0150]** The case **110** may include a board accommodating portion **113** for accommodating the circuit board **150**. More specifically, the circuit board **150** and a bracket panel **130** on which the circuit board **150** is mounted may be accommodated in the board accommodating portion **113** of the case **110**.

**[0151]** The case **110** may include a grill installing portion **112** extending upward from the board accommodating portion **113** and protruding forward from the board accommodating portion **113**.

**[0152]** The grill installing portion **112** may form a portion of the cooling flow path C. More specifically, outside air may enter from above the grill installing portion **112** and flow downward from the grill installing portion **112**. The opening **111** of the case **110** may be formed in an upper portion of the grill installing portion **112**.

**[0153]** In other words, a front surface of the grill installing portion **112** may protrude forward from a front surface of the board accommodating portion **112**. Also, a portion of the front surface of the grill installing portion **112** may extend in an up-down direction to enlarge an inflow space which outside air enters.

**[0154]** Accordingly, because the grill installing portion **112** protrudes forward from the board accommodating portion **113** of the case **110**, an inlet area of the cooling flow path C may further increase in a front direction. Also, because the grill installing portion **112** extends upward from the board accommodating portion **113**, the inlet area of the cooling flow path C may further increase in the up-down direction. That is, because an entire area through which cooling air enters increases, cooling efficiency of the cooking apparatus **1** may be raised.

**[0155]** The suction grill **120** may be coupled with the upper portion of the case **110** and include the first cooling air inlet **123** formed in a top of the case **110**. The suction grill **120** may be screw-coupled with the case **110**. This will be described in detail, below.

**[0156]** The control unit **100** may include the circuit board **150**, the bracket panel **130** on which the circuit board **150** is mounted, and a guide member **140** installed on an upper portion of the bracket panel **130**.

**[0157]** The bracket panel **130** may be accommodated inside the case **110**. The bracket panel **130** may be substantially in a shape of a box of which the rear side opens. The circuit board **150** may be accommodated inside the bracket panel **130**. The circuit board **150** may be fixed inside the bracket panel **130**.

**[0158]** The bracket panel **130** may protect the circuit board **150** from an external impact or a foreign material.

**[0159]** The circuit board **150** may be accommodated in a rear side of the bracket panel **130** and positioned at a communicating portion **84** of the front plate **80**. In other words, the circuit board **150** may communicate with the first machine room **61**. Accordingly, cooling air may enter the suction grill **120** to cool the circuit board **150** while cooling components of the first machine room **61**.

**[0160]** The guide member **140** may be installed on the upper portion of the bracket panel **130**.

**[0161]** A preventing portion may be provided to prevent water or a foreign material from entering the circuit board **150** through the first cooling air inlet **123** of the suction grill **120**.

**[0162]** The preventing portion may enable water collected on an upper surface of the preventing portion to be easily discharged to the outside of the control unit **100**. The guide member **140** may be positioned below the second cooling air inlet **83** of the front plate **80**.

**[0163]** FIG. **11** is a rear perspective view of the door shown in FIG. **9**.

**[0164]** As shown in FIGS. **9** to **11**, the door **20** may close and open the opening **80***a* of the front plate **80** and cover a front portion of the case **110**. A portion of the rear surface of the door **20** may be depressed inward to accommodate the case **110**.

**[0165]** The door **20** may include a sealing portion **21** for opening and closing the opening **80***a* in front of the cooking room **30**.

**[0166]** The sealing portion **21** may be positioned in front of the main body **10**. The sealing **21** may include a shielding member (not shown) to prevent a high frequency existing inside the cooking room **30** from leaking out. A portion of the sealing portion **21** may be transparent to enable a user to check the inside of the cooking room **30** from the outside.

**[0167]** The sealing portion **21** may include the door latch **211** inserted in the door fixing portion **82** of the front plate **80**. The door latch **211** may extend from a rear surface of the sealing portion **211** toward the main body **10** and be formed at a location corresponding to the door fixing portion **82**.

**[0168]** One side of the sealing portion **21** may be hinge-coupled with the main body **10** such that the door **20** is rotatable with respect to the main body **10**.

**[0169]** The door **20** may include a cover portion **22** extending from a side portion of the sealing portion **21** and covering the front portion of the case **110**.

**[0170]** The cover portion **22** may be formed by being depressed inward from the rear surface of the door **20**. More specifically, the cover portion **22** may have a smaller thickness than the sealing portion **21**. Accordingly, the control unit **100** may be accommodated behind the cover portion **22**. That is, the control unit **100** may be positioned to one side of the sealing portion **21**.

**[0171]** The cover portion **22** may include a first cover portion **221** and a second cover portion **222**.

**[0172]** The first cover portion **221** may be depressed forward from the rear surface of the door **20** to cover the front surface of the grill installing portion **112** of the case **110**.

**[0173]** The second cover portion **222** may be depressed forward from the rear surface of the door **20** to cover a portion of the front surface of the board accommodating portion **113** of the case **110**.

**[0174]** The first cover portion **221** may be further depressed forward than the second cover portion **222**. Because the grill installing portion **112** of the case **110** protrudes forward from the board accommodating portion **113**, the rear surface of the door **20** may also correspond to the grill installing portion **112**.

**[0175]** The door **20** may include the handle portion **23** that is spaced from the front portion of the control unit **100** to form a grip space.

**[0176]** More specifically, the handle portion **23** may be formed at a lower area of the first cover portion **221**. The handle portion **23** may be positioned to one side of the second cover portion **222**.

**[0177]** The handle portion **23** may be provided on the same plane as the first cover portion **221**. More specifically, the handle portion **23** may be not stepped to the first cover portion **221**. Accordingly, the handle portion **23** may be further depressed forward from the rear surface of the door **20** than the second cover portion **222**, like the first cover portion **221**.

**[0178]** FIG. **12** is an exploded view showing a state in which a control unit of a cooking apparatus according to an embodiment of the disclosure is covered by a door.

**[0179]** Referring to FIG. **12**, the control unit **100** may be positioned between the door **20** and the main body **10**. More specifically, the control unit **100** may be positioned behind the cover portion **22** of the door **20**. Also, the control unit **100** may be positioned to one side of the sealing portion **21** of the door **20**.

**[0180]** The suction grill **120** of the control unit **100** may be aligned with the sealing portion **21** at one side of the sealing portion **21**.

**[0181]** Also, the discharge panel **200** in which the second circulating air outlet **210** is formed may be formed on the top of the main body **10**. More specifically, the discharge panel **200** may be installed on the external housing **11** to be positioned behind the sealing portion **21** of the door **20**. In other words, the sealing portion **21** may be positioned in front of the circulating air outlet.

**[0182]** The first cover portion **221** of the door **20** may cover the front surface of the grill installing portion **112** of the case **110**. The second cover portion **222** of the door **20** may cover the front surface of the board accommodating portion **113** of the case **110**. The grip space may be formed between the handle portion **23** provided to one side of the second cover portion **222** of the door **20** and the front surface of the board accommodating portion **113** of the case **110**.

**[0183]** FIG. **13** shows a state before a suction grill of a cooking apparatus according to an embodiment of the disclosure is installed in a case. FIG. **14** shows a state after the suction grill of FIG. **13** is installed in the case.

**[0184]** Referring to FIGS. **13** and **14**, the case **110** of the control unit **100** may include a coupling boss **114**.

**[0185]** The coupling boss **114** may be formed in an inner side of the grill installing portion **112** of the case **110**. The coupling boss **114** may couple the case **110** with the suction grill **120**.

**[0186]** The suction grill **120** may include a grill portion **121** and a coupling portion **122**.

**[0187]** In the grill portion **121**, the first cooling air inlet **123** may be formed. The coupling portion **122** may be positioned to one side of the grill portion **121**. A coupling hole **124** for coupling with the case **110** may be formed in the coupling portion **122**.

**[0188]** The suction grill **120** may be detachable from the case **110** to form an entrance of the cooling flow path C.

**[0189]** Air entering the first cooling air inlet **123** of the suction grill **120** may enter the inside of the main body **10** through the second cooling air inlet **83** of the front plate **80**.

**[0190]** The first cooling air inlet **123** of the suction grill **120** may be biased with respect to the second circulating air outlet **210** of the discharge panel **200**. More specifically, because the first cooling air inlet **123** is provided in front of the main body **10**, and the second circulating air outlet **210** is provided in the top of the main body **10**, the first cooling air inlet **123** may be biased with respect to the second circulating air outlet **210** in a front-back direction of the cooking apparatus **1**.

**[0191]** Also, because the first cooling air inlet **123** is provided to one side of the sealing portion **21** of the door **20** and the second circulating air outlet **210** is provided behind the sealing portion **21**, the first cooling air inlet **123** may be biased with respect to the second circulating air outlet **210** in a left-right direction of the cooking apparatus **1**. In other words, because the first cooling air inlet **123** is provided behind the cover portion **22** of the door **20** and the second circulating air outlet **210** is provided behind the sealing portion **21** of the door **20**, the first cooling air inlet **123** may be biased with respect to the second circulating air outlet **210** in the left-right direction of the cooking apparatus **1**.

**[0192]** Accordingly, because the first cooling air inlet **123** forming an entrance of the cooling flow path C and the second circulating air outlet **210** forming an exit of the circulating flow path are biased with respect to each other, a phenomenon in which heated air discharged from the circulating flow path enters the cooling flow path C may be reduced. Accordingly, the entire cooling efficiency of the cooking apparatus **1** may be improved.

**[0193]** FIG. **15** is a perspective view of a cooking apparatus according to an embodiment of the disclosure after an external housing is removed from the cooking apparatus. FIG. **16** is a cross-sectional view taken along line A-A' of FIG. **15**. FIG. **17** is an enlarged view of a upper-front portion of FIG. **16**.

**[0194]** Referring to FIGS. **15** to **17**, a flow of air flowing along the cooling flow path C will be described in detail.

**[0195]** First, outside air may enter the inside of the control unit **100** through the first cooling air inlet **123** formed in the upper portion of the control unit **100**.

**[0196]** The air entering the inside of the control unit **100** may flow to the first machine room **61** through the second cooling air inlet **83** (see FIG. **13**) of the front plate **80**.

**[0197]** The air entering the first machine room **61** may cool the rear surface of the control unit **100** and the inside of the first machine room **61** and then flow to the second machine room **62** through the cooling fan **71**.

**[0198]** The air entering the second machine room **62** may cool the magnetron **73** and the transformer **74** and then flow to the guide duct **72** while being prevented from flowing to the second circulating fan **42** by the cooling divider **51**.

**[0199]** The air entering the guide duct **72** may flow to the cooking room **30** and cool the rear surface of the sealing portion **21** of the door **20**. After the air cools the rear surface of the sealing portion **21** of the door **20**, the air may flow to the exhaust duct **31** communicating with the cooking room **30** at the upper surface.

**[0200]** The air entering the exhaust duct **31** may be discharged to the outside of the cooking apparatus **1** through an exhaust portion **311** and the cooling air outlet **93** (see FIG. **93**) of the external housing **11**. The above-described path may be the cooling flow path C of the cooking apparatus **1** according to an embodiment of the disclosure.

**[0201]** Referring to FIG. **16**, the first cover portion **221** and the second cover portion **222** of the door **20** may be shaped to correspond to the grill installing portion **112** and the board accommodating portion **113** of the case **110** of the control unit **100**. More specifically, because the grill installing portion **112** protrudes forward from the board accommodating portion **113**, the first cover portion **221** of the door **20** may also be further depressed forward than the second cover portion **222**.

**[0202]** Accordingly, the door **20** of the cooking apparatus **1** according to an embodiment of the disclosure may form a wider entrance of the cooling flow path C by adjusting the thickness of the cover portion **22** while maintaining a constant thickness of the sealing portion **21**. Accordingly, slimness of the door **20** may be realized.

**[0203]** More specifically, by enlarging the grill installing portion **112** of the case **110**, a size of the suction grill **120** may increase, thereby securing a greater quantity of air entering the cooling flow path C.

**[0204]** Referring to FIG. **17**, the suction grill **120** according to an embodiment of the disclosure may be inclined in the front direction. More specifically, an inclination angle a of the suction grill **120** with respect to a horizontal reference line may be about **5** degrees or less.

**[0205]** In a case in which the inclination angle a of the suction grill **120** is too great, a height of the door **20** may increase correspondingly although a quantity of air entering the cooling flow path C increases.

**[0206]** Also, in a case in which the inclination angle a of the suction grill **120** is too small, a sufficient quantity of air entering the cooling flow path C may be not secured.

**[0207]** In the cooking apparatus **1** according to an embodiment of the disclosure, cooling air may enter through the suction grill **120** positioned to one side of the sealing portion **21** of the door **20** and accordingly, a sufficient width in the left-right direction may be not secured. Accordingly, by forming the suction grill **120** inclined in the front direction, a greater inflow quantity of cooling air may be secured with a limited width.

**[0208]** FIG. **18** is an enlarged cross-sectional view showing a right side of a cooking apparatus according to an embodiment of the disclosure.

**[0209]** As shown in FIG. **18**, in an outer side of the door **20** of the cooking apparatus **1** according to an embodiment of the disclosure, the handle portion **23** may be formed. More specifically, the handle portion **23** may be formed at a side portion of the second cover portion **222**.

**[0210]** Also, referring to FIG. **18**, the second machine room **62** may be formed behind the first machine room **61**, and the second circulating fan **42** may be positioned behind the second machine room **62**.

**[0211]** Because the second circulating fan **42** is positioned behind the second machine room **62**, a height in up-down direction of the cooking apparatus **1** may be reduced. Accordingly, the cooking apparatus **1** having a slimmer height may be implemented.

**[0212]** In other words, as shown in FIGS. **7** and **18**, the machine room **60** and the second circulating fan **42** may be positioned to a right side of the cooking room **30**, and the first circulating fan **41** may be positioned to a left side of the cooking room **30**. Accordingly, the cooking apparatus **1** having a slimmer height in the up-down direction may be implemented.

**[0213]** Also, because the first cooling air inlet **123** is formed in the top of the cooking apparatus **1**, and all of the cooling air outlet **93** and the first and second circulating air outlets **911** and **210** are formed in the top of the cooking apparatus **1**, a height required for an air inlet or an air outlet may be reduced.

**[0214]** That is, the cooking apparatus **1** according to an embodiment of the disclosure may have a slim thickness in the up-down direction.

**[0215]** Also, by arranging an entrance of the cooling flow path C and exits of the circulating flow paths V**1** and V**2** to be maximally away from each other, cooling efficiency of the cooking apparatus **1** may be more improved.

**[0216]** By positioning the cooling air inlet of the cooling flow path and the circulating air outlet of the circulating flow path to be maximally away from each other, the cooling efficiency of the cooking apparatus may be improved.

**[0217]** By forming the inclined suction grill, the door may have a slim thickness and the air inflow area of the cooling flow path may increase.

**[0218]** Although a few embodiments of the disclosure have been shown and described, it would be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.

What is claimed is:

**1**. A cooking apparatus comprising:

a main body forming a cooking room therein and including a front plate with an opening to the cooking room;

a case positioned to cover one side of the front plate and formed to accommodate a circuit board therein;

a door opening and closing the opening;
a circulating air outlet formed at a top of the main body, and configured to filter and discharge air received from below the main body; and
a suction grill mounted on an upper portion of the case, wherein a cooling air inlet, which allows cooling air to enter through the suction grill, is positioned in front of the circulating air outlet.

**2**. The cooking apparatus of claim **1**, wherein the front plate comprises an installing portion positioned to one side of the opening and on which the case is mounted.

**3**. The cooking apparatus of claim **1**, wherein the suction grill is detachable from the case.

**4**. The cooking apparatus of claim **1**, further comprising:
a discharge panel attachable to and detachable from the main body, wherein the circulating air outlet is formed in the discharge panel, and the discharge panel is positioned behind the door.

**5**. The cooking apparatus of claim **1**, wherein the case comprises:
a board accommodating portion formed to accommodate the circuit board; and
a grill installing portion formed to extend upward from the board accommodating portion and protrude forward from the board accommodating portion.

**6**. The cooking apparatus of claim **1**, wherein the suction grill is inclined along a front direction of the door.

**7**. The cooking apparatus of claim **1**, wherein the cooling air inlet is a first cooling air inlet, and
the front plate includes a second cooling air inlet provided behind the first cooling air inlet to allow air entering the first cooling air inlet to flow to an inside of the main body.

**8**. The cooking apparatus of claim **1**, wherein the door comprises:
a sealing portion opening and closing the opening in front of the cooking room; and
a cover portion positioned to one side of the sealing portion and to cover the front portion of the case, wherein a thickness of the cover portion is smaller than a thickness of the sealing portion.

**9**. The cooking apparatus of claim **1**, further comprising:
a machine room provided behind the front plate where air entering through the suction grill enters the machine room; and
a guide duct to guide the air entering the machine room to the cooking room.

**10**. The cooking apparatus of claim **1**, wherein the suction grill comprises:
a grill portion in which the cooling air inlet is formed; and
a coupling portion positioned to one side of the grill portion and coupleable with the case.

**11**. The cooking apparatus of claim **1**, further comprising:
an internal housing forming the cooking room;
an external housing positioned outside the internal housing and forming an outer appearance of the cooking apparatus;
at least one circulating fan positioned between the internal housing and the external housing, and configured to suck polluted air from below the main body; and
a partition bracket forming a flow path along which air drawn by the at least one circulating fan and entering an inside of the main body flowing, the partition bracket positioned behind the internal housing and formed to extend horizontally between the internal housing and the external housing.

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